

GeRe: Towards Efficient Anti-Forgetting in Continual Learning of LLM via General Samples Replay

Yunan Zhang, Shuoran Jiang, Mengchen Zhao, Yuefeng Li, Yang Fan, Xiangping Wu, Qingcai Chen

Abstract—The continual learning capability of large language models (LLMs) is crucial for advancing artificial general intelligence. However, continual fine-tuning LLMs across various domains often suffers from catastrophic forgetting, characterized by: 1) significant forgetting of their general capabilities, and 2) sharp performance declines in previously learned tasks. To simultaneously address both issues in a simple yet stable manner, we propose General Sample Replay (GeRe), a framework that use usual pretraining texts for efficient anti-forgetting. Beyond revisiting the most prevalent replay-based practices under GeRe, we further leverage neural states to introduce a enhanced activation states constrained optimization method using threshold-based margin (TM) loss, which maintains activation state consistency during replay learning. We are the first to validate that a small, fixed set of pre-collected general replay samples is sufficient to resolve both concerns—retaining general capabilities while promoting overall performance across sequential tasks. Indeed, the former can inherently facilitate the latter. Through controlled experiments, we systematically compare TM with different replay strategies under the GeRe framework, including vanilla label fitting, logit imitation via KL divergence and feature imitation via L1/L2 losses. Results demonstrate that TM consistently improves performance and exhibits better robustness. Our work paves the way for efficient replay of LLMs for the future. Our code and data are available at <https://github.com/Qznan/GeRe>.

Index Terms—Large Language Models, Continual Learning, Finetune, Replay, Activation State

1 INTRODUCTION

CONTINUAL learning (CL) of large language models (LLMs) remains challenging for real-world applications. For instance, continual finetuning often degrades general capabilities, particularly over long task sequences. The finetuned model forgets its original world knowledge or basic instruction-following skills [1], [2]. Additionally, the overall performance on sequential downstream tasks often deteriorates due to forgetting of previously learned

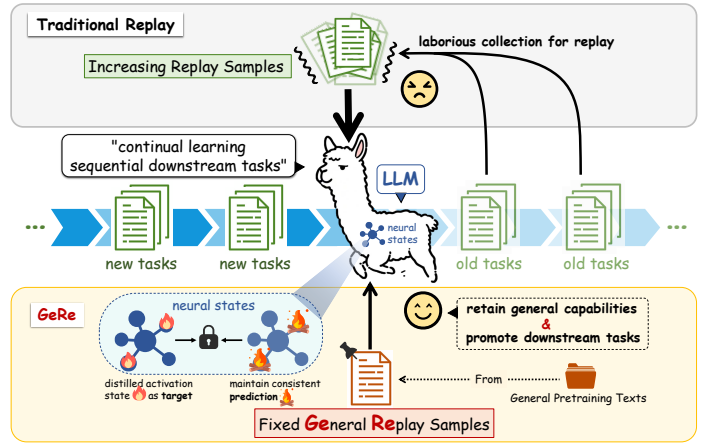


Fig. 1: Traditional replay vs. GeRe: unlike traditional replay requiring laborious collection of an increasing set of downstream replay samples, GeRe simply employs a fixed set of general replay samples to not only retain general capabilities in continual learning, but also enhance the overall performance of learned downstream tasks. The blue oval is the threshold-based margin loss that imposes consistency constraint on neural activation state under GeRe frameworks.

tasks, caused by inter-task conflicts. This phenomenon, also known as catastrophic forgetting, often compels practitioners to seek complex CL solutions. However, the contemporary LLM system, marked by architectural bulkiness and computational heaviness, is imperative to call for a simple yet stable approach to effectively mitigate forgetting. In this context, our research aims to review and develop an efficient and general anti-forgetting method of CL adapted to the LLM era.

Historically, solutions for CL are primarily categorized into three traditional branches: replay-based, regularization-based, and architecture-based methods [3]. Considering the massive number of parameters in LLMs and their widely accepted fixed structures, it appears prohibitive and impractical to regularize all parameters or frequently expand the architecture for every new task. Thus current practice in LLMs continual learning regularly prioritizes replay-based methods due to its simplicity. For instance, practitioners commonly mix a certain proportion of general task samples

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during finetuning for downstream tasks [4], [5]. However, the underlying mechanisms and optimal strategies of these replay-based methods tailored for LLMs remain insufficiently explored and analyzed.

In this work, we (1) systematically revisit the replay mechanisms targeting LLMs under a newly introduced general sample replay (GeRe) framework and, (2) present a threshold-based margin (TM) loss for activation state constrained optimization. Specifically, we prepare a fixed, permanently reusable set of general replay samples (e.g., the commonly used pretraining texts) and leverage the TM loss to maintain consistent neuron activation states, ultimately resisting various forms of forgetting.

The approach is motivated by two ideas: (1) From a cognitive perspective, a learner obtaining superior general capabilities is more likely to achieve better generalization and robustness in downstream tasks. Leveraging its comprehensive knowledge, such a learner can reduce conflicts arising from overfitting to specific tasks, thereby mitigating task forgetting. Consequently, it is worth exploring how to utilize general replay samples to retain general capabilities. (2) In the human brain, critical information is sparsely distributed across a few activated neurons [6], [7]. Therefore, in replay-based continual learning (replay learning), the activation states of neurons evoked by replay samples may require deeper attention. By designing an activation state constrained optimization, we seek a less rigid but more informative target that enables the replay learning to be more robust and generalizable.

Through the paper we have explored and answered 2 pressing questions in real-world LLMs continual learning scenario:

Q1: Can we simply select a fixed set of replay samples once and for all? To retain general capabilities, contemporary strategies for mixing replay samples in LLM training may be as laborious as feature engineering, requiring careful selection of both the proper size and specific replay samples tailored to the particular downstream task. For instance, even with a fixed mixing ratio, we still need to frequently re-size the replay samples set and select an appropriate subset or superset to adapt to the varying data scale of incoming tasks. For this question, we have empirically validated that constructing a fixed set of randomly selected general replay samples (e.g., 1k texts from the widely available general pretraining corpus) can be durably applied to fulfill all replay needs in subsequent tasks, while successfully preserving general capabilities. This becomes more pronounced when integrating replay with feature-based distillation, as it fully exploits information from these limited replay samples rather than merely fitting their explicit labels. To our knowledge, we are the first to propose that a fixed set of general replay samples can efficiently adapt to real-world continual learning scenario involving long sequences of tasks under full or LoRA settings, which holds significant practical implications.

Q2: Can general replay samples alone facilitate continual learning in sequential downstream tasks, typically without any of task replay samples? Normally, collecting task replay samples from each old task in subsequent learning is necessary to maintain their long-term performance. However, we believe that the learning efficacy of any down-

stream task fundamentally depends on the LLM’s general knowledge. For this question, we have encouragingly validated that the aforementioned fixed set of general replay samples, under our optimization approach, can effectively promote the persistent retention of previously learned task knowledge, mitigating the forgetting induced by inter-task conflicts. The results demonstrate the feasibility of conveniently utilizing only predetermined general replay samples to resist task-specific forgetting in future applications.

These answers highlight the advantages of the proposed GeRe framework. Furthermore, we enhance feature-based replay learning under GeRe by introducing activation state constrained optimization, which statistically determines activation states and optimizes using a threshold-based margin loss. This relatively lightweight constraint on feature values empirically exhibits better robustness and generalizability compared to the conventional yet rigid L1/L2 fitting manner.

Our contributions are as follows:

- **1. GeRe:** We first demonstrate that a fixed set of predefined general replay samples can be reused throughout the entire continual finetuning process, effectively preserving LLM’s original general capabilities. Crucially, replaying any downstream task sample proves unnecessary, as maintaining general capabilities alone enhances overall downstream tasks performance.
- **2. TM loss:** We pioneer a comprehensive comparison of commonly used replay-based practices for continual learning in LLMs, exploring their integration with various knowledge distillation strategies. Among these, our proposed threshold-based margin loss, motivated by the previously overlooked activation state constraint, achieves SoTA performance.
- **3.** Our method shows robustness not only to learning rate—a critical hyperparameter seriously impacting both knowledge updating and retention—but also to intrinsic optimization dynamics, as evidenced by optimization landscape visualization, highlighting its practical utility.

2 RELATED WORKS

2.1 Continual Learning

Continual learning, also known as lifelong or incremental learning, refers to the ability of a machine learning model to learn from a stream of data over time, while retaining knowledge from previous tasks and adapting to new ones without forgetting [8]. Unlike traditional learning paradigms, where models are trained on static datasets, continual learning should address the dynamic nature of real-world applications, where data distributions and tasks evolve over time.

A central challenge in continual learning is the catastrophic forgetting problem, where a model tends to forget previously learned knowledge when trained on new tasks [9]. To mitigate this, various strategies have been proposed, including replay-based, regularization-based, and architecture-based methods. For instance, replay-based methods like Experience Replay [10] store and replay samples from past tasks to maintain performance, while regularization-based methods like Elastic Weight Consolidation (EWC) [11] introduce a regularization term to preserve

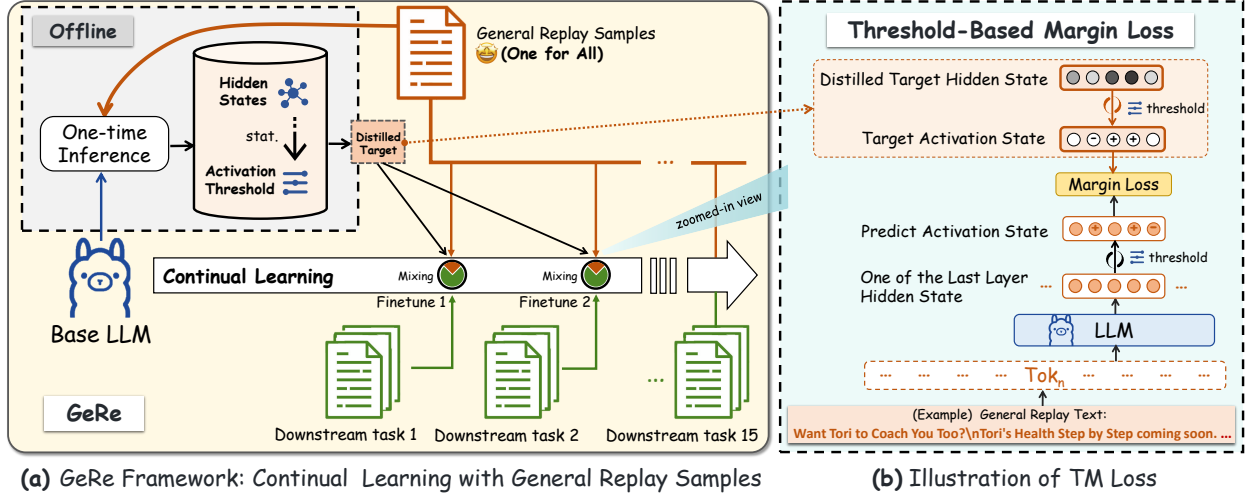


Fig. 2: (a) Flowchart of the GeRe framework using general replay samples, including distillation of hidden states and the derived activation state in offline mode, and continual learning across sequential tasks with mixing general samples for replay. (b) Illustration of threshold-based margin loss, which transforms the hidden values into discrete activation states on both the target and prediction followed by margin loss calculation.

important parameters for previous tasks. Similarly, Learning without Forgetting (LwF) [12] uses knowledge distillation to regularize the model by minimizing the divergence between its current and previous outputs. Architecture-based methods allocate distinct subsets of model parameters to different tasks to prevent interference. For example, Progressive Neural Networks (PNNs) [13] expand the network architecture by adding new layer of parameters for each task while freezing existing ones. In addition Mask-Based Methods [14] learns task-specific masks to trigger or suppress parameters dynamically.

In the LLM era, continual pretraining or finetuning has become essential for model iteration and advancement. We focus on continual finetuning, where the most common practice involves replay-based methods, which typically incorporate general corpus to preserve the model’s general capabilities for downstream tasks. In this work, we explore a further integration with regularization-based techniques, i.e., the distillation strategy used in LwF. Specifically, we pre-generate and store the feature representations of the replay samples, which are then used as targets within a distillation framework to effectively leverage this information. Moreover, we include the LoRA [15] setup since it is widely adopted in finetuning due to its strong generalization and resistance to forgetting. It can be considered as another type of architecture-based method, as it typically trains only a small fraction of parameters.

2.2 Knowledge Distillation

Knowledge distillation (KD) [16] aim to compress large models into smaller, efficient ones by transferring knowledge from a teacher model to a student model. This process is achieved by minimizing the difference between their output distributions, where the student learns from the teacher’s soft labels rather than the original dataset’s hard labels [17], [18], [19]. KD can be implemented generally with two types: logit-based imitation and feature-based imitation [20]. The former involves matching the predictions and

target distributions using the Kullback-Leibler divergence (KL) loss with a temperature-based softmax normalization, while the latter focuses on aligning the intermediate representations in the feature space through similarity-based functions.

KD has already been applied in continual learning, with the key distinction lying in how the target and prediction are defined. Taking LwF for example, when the new task’s samples arrived, the model preliminarily computed the logits of these samples at the output heads regarding old tasks, served as the distilled pseudo-targets. During subsequent training, the real-time predicted logits at these old tasks output heads are constrained to match the pre-computed pseudo-targets. In this case, the teacher and student models are essentially the same model. This self-distillation mechanism [21] enables the model to retain performance on previously learned tasks while adapting to new ones.

However, few research emphasizes similarity of feature in KD typically applied for continual learning of LLMs [22]. Our work thereby delves deeper into studying efficient mechanism combining replay and distillation methods, where labels or features of replay samples are pre-distilled to serve as pseudo-targets, enabling persistent fitting of replay samples during continual learning. We empirically compare the effect of using replay samples simply versus leveraging replay samples under both logit-based imitation via KL divergence and feature-based imitation via L1 or L2 function. The study offers a comprehensive comparison of diverse replay strategies.

3 PROPOSED METHOD

This section provides a detailed introduction to the overall process of GeRe framework and the proposed replay-based activation state constrained optimization. As outlined in Fig.2(a), we first collect a small-scale set of general samples for permanently available replay. Then these data are proactively distilled using the untuned base LLM to derive

the activation threshold, which determines the activation state. Subsequently, continual finetuning is performed on a mixed data containing downstream task and general replay samples, jointly optimizing a specialized replay-based objective alongside the standard cross-entropy loss. Fig. 2(b) illustrates our proposed threshold-based margin (TM) loss, which transforms the given optimization target into activate states through thresholds on both sides and employs a margin loss for constraint. (The different optimization targets used by other competitors are shown in Fig. 3.)

3.1 Distilled Activation States

In deep neural networks, neuron activation values refer to the outputs of each layer’s sub-network. Taking the Transformer-based LLMs as an example, the activation values refer to the output of the feed-forward network within each layer. These activations are progressively passed through residual connections, evolving started from the input and ultimately forming the network’s final output.

Analogous to the activation states of neurons in the human brain, we propose to categorize the neural network activations into three distinct states: positive activation, negative activation, and non-activation. These states exhibit discrete sparsity patterns while encoding specific semantic information. Building on this, we hypothesize that during continual finetuning, the original activation states represented with general replay samples can effectively reflect the model’s general capabilities. Therefore, in our replay learning, we employ feature-based imitation to utilize these activations as targets, thereby preserving the essential characteristics of the model’s learned representations.

3.1.1 Feature-Based Distillation

Given a general replay sample set $\mathcal{D}^{(g)} = \{s_1, s_2, \dots, s_N\}$ comprising N natural sentences s , we feed these sentence samples into the base LLM, performing forward propagation to obtain the activation values (i.e., the hidden states output of each layer) as follows:

$$\bar{\mathbf{h}} = \text{LLM}(s) \quad (1)$$

where $\bar{\mathbf{h}} \in \mathbb{R}^{n^t \times n^d \times L}$ is activation value tensor, n^t is the length (number of tokens) of the input, n^d is the dimension of the hidden states, L is the number of layers in LLM. We distill these feature-base activation values of all samples in $\mathcal{D}^{(g)}$ to form $\mathcal{H}^{(g)} = \{\bar{\mathbf{h}}^1, \bar{\mathbf{h}}^2, \dots, \bar{\mathbf{h}}^N\}$.

3.1.2 Activation Threshold

After distilling all the activation values of samples in $\mathcal{D}^{(g)}$, we statistically determine the activation threshold and accordingly infer the activation state. Specifically, for each activation value $\bar{h}_{j,k,l}$ corresponding to the k -th dimension of hidden state at the l -th layer, we compute its mean and variance across the entire $\mathcal{H}^{(g)}$ over the size N and length n^t , yielding $mean_l = (m_1, m_2, m_k, \dots, m_{n^d})_l$ and $std_l = (\sigma_1, \sigma_2, \sigma_k, \dots, \sigma_{n^d})_l$ relative to the l -th layer. In practice, since the hidden state of the last layer encodes the majority of the semantic information for the model’s final predictions, we choose to utilize only the last layer for subsequent computations, which serve as the constraint optimization objective. Therefore, we assume $l = L$ (the

last layer) and omit the subscript l in all the following formulas, (e.g., $\bar{h}_{j,k} := \bar{h}_{j,k,l=L}$). Each component m_k and σ_k is computed as follows:

$$m_k = \frac{1}{N \times n^t} \sum_{i=1}^N \sum_{j=1}^{n^t} \bar{h}_{j,k}^i \quad (2)$$

$$\sigma_k = \sqrt{\frac{1}{N \times n^t} \sum_{i=1}^N \sum_{j=1}^{n^t} (\bar{h}_{j,k}^i - m_k)^2} \quad (3)$$

where i ranges over the $\mathcal{D}^{(g)}$ size N and j ranges over the number of tokens within the current sample, k denote the k -th dimension. We further utilize the characteristics of Gaussian distribution to define the activation thresholds, considering one standard deviation above the mean as the positive activation threshold: $\tau^+ = m + 1\sigma$, and one standard deviation below the mean as the negative activation threshold: $\tau^- = m - 1\sigma$. Each $\bar{h}_k$ relative to the k -th dimension possesses two thresholds as follows:

$$\tau_k = (\tau_k^-, \tau_k^+) = (m_k - 1 \cdot \sigma_k, m_k + 1 \cdot \sigma_k) \quad (4)$$

and we define three types of activation state: 1) values greater than τ^+ are considered positively activated, 2) values less than τ^- are considered negatively activated, 3) values between τ^- and τ^+ are considered non-activated:

$$\text{state}_k = \begin{cases} \text{positively activated} & \text{if value} < \tau_k^- \\ \text{non-activated} & \text{if } \tau_k^- \leq \text{value} \leq \tau_k^+ \\ \text{negatively activated} & \text{if value} > \tau_k^+ \end{cases} \quad (5)$$

According to Gaussian distribution, about 68.27% of the activation values are considered non-activated, which aligns with the assumption that only a subset of neurons plays a critical role during forward propagation.

Once the thresholds for the $\mathcal{D}^{(g)}$ are determined, they can be permanently applied to subsequent downstream task finetuning conveniently. Notably, we can also preemptively transform the float-type activation values into binary-type activation states to reduce the storage overhead.

3.2 Threshold-Based Margin Optimization

This section describes the computation process of the proposed TM loss. Optionally we can randomly select a subset of samples from $\mathcal{D}^{(g)}$ for actually replay. However, if the original size is small, using the complete set is recommended. Specifically, given the previously determined positive and negative activation thresholds, these samples are jointly optimized with the downstream task samples during continual finetuning. Detailed steps are as follows.

3.2.1 Batch Insertion

Traditional replay-based methods simply mix replay samples with downstream task training samples randomly. However, it requires considering the scale of both samples set and thereby adjusting the mixing ratio. Additionally, during optimization, it is possible that a given batch may contain no replay samples, resulting in gradients that are exclusively influenced by the downstream task samples. To address this issue, we propose the Batch Insertion (BI) strategy, which ensures that a specific proportion of replay

samples is included in each training batch. This strategy encourages the influence of the gradient update direction by the replay samples in every iteration, helping to retain the general capabilities of the LLM, meanwhile avoids the cumbersome adjustment of mixing ratios for datasets of varying scales.

Given the finetuning batch size n^{batch} , we define the Batch Insertion ratio as ρ^{BI} , indicating that $\rho^{\text{BI}} \times n^{\text{batch}}$ samples in each batch are replay samples. This can be easily implemented by modifying the *Sampler Class* within *Torch DataLoader*.

3.2.2 Loss Calculation

During the joint training process, the proposed TM loss for the replay samples within each batch is computed as follows:

$$\mathcal{L}_{j,k}^{\text{TM}} = \begin{cases} \max(\hat{h}^{j,k} - \tau_k^-, 0) & \text{if } \bar{h}^{j,k} < \tau_k^- \\ \max(\hat{h}^{j,k} - \tau_k^+, 0) & \text{if } \tau_k^- \leq \bar{h}^{j,k} \leq \tau_k^+ \\ + \max(\tau_k^- - \hat{h}^{j,k}, 0) & \\ \max(\tau_k^+ - \hat{h}^{j,k}, 0) & \text{if } \bar{h}^{j,k} > \tau_k^+ \end{cases} \quad (6)$$

where $\mathcal{L}_{j,k}^{\text{TM}}$ denotes the TM loss for the k -th dimension of the hidden state on the j -th token (at the last layer), $\bar{h}$ is the pre-computed target activation values while $\hat{h}$ is the currently predicted activation values. This piecewise loss function guides the optimization direction of the predicted value when the target value resides in the negatively activated, non-activated, and positively activated states, respectively. The overall TM loss for each replay sample sentence is computed as follows:

$$\mathcal{L}^{\text{TM}} = \frac{1}{n^t \times n^d} \sum_{j=1}^{n^t} \sum_{k=1}^{n^d} \mathcal{L}_{j,k}^{\text{TM}} \quad (7)$$

3.2.3 Dynamic Weight Balancing

During training, we jointly optimize the TM loss $\mathcal{L}^{\text{TM}}$ regarding general replay samples and the standard Cross-Entropy (CE) loss $\mathcal{L}^{\text{CE}}$ regarding downstream task samples. To prevent the model from being overly biased toward optimizing either loss, we adopt a dynamic loss weighting strategy to balance their magnitudes as follows:

$$\omega^{\text{TM}} = \text{detach}(\mathcal{L}^{\text{CE}} / \mathcal{L}^{\text{TM}}) \quad (8)$$

$$\mathcal{L} = \mathcal{L}^{\text{CE}} + \omega^{\text{TM}} \cdot \mathcal{L}^{\text{TM}} \quad (9)$$

where $\mathcal{L}$ denotes the final total loss for continual finetuning. ω^{TM} is the dynamic weight to dynamically scale the magnitude of the TM loss to match that of the CE loss during joint optimization. The $\text{detach}()$ function indicates that the weight value is detached from gradient backpropagation, preventing it from being optimized. Notably, this approach is experimental, employing fixed or dynamic weights depending on practice.

4 EXPERIMENTS

In this section, we evaluate the performance of our proposed method using a representative base-LLM Llama-3.1-8B [23] along with 15 downstream tasks under continue

learning regime. We first introduce the datasets, metrics and experimental settings, followed by detailed analyses of the experimental results and landscape visualization [24], [25] exploring robustness.

4.1 Datasets

For the general replay sample set $\mathcal{D}^{(\text{g})}$, we randomly select 1K samples from the open-source SlimPajama-627B corpus [26], which is a cleaned and deduplicated version of RedPajama [27] that reproduces the collection of LLaMA training data. We release the complete selected samples used throughout this paper to ensure reproducibility. Notably, the selection process is arbitrary rather than deliberately curated (see Appendix for details), which further substantiating the robustness and universality of our method with respect to the replay data. This replay data potentially reflect the general ability of the base LLM model, which is used to calculate the activation threshold and to compute the threshold-based margin loss.

For the downstream finetuning tasks, we adopt a long-sequence continual learning benchmark comprising as many as 15 diverse datasets [28], which enables a comprehensive evaluation of model performance in practical scenarios under more demanding and challenging conditions. The benchmark integrates 5 datasets (yelp, amazon, dbpedia, agnews, yahoo) from the standard CL benchmark [29], [30], 4 datasets (MNLI, QQP, RTE, SST-2) from the GLUE benchmark [31], 5 datasets (CB, COPA, MultiRC, BoolQA, WiC) from the SuperGLUE benchmark [32], and the IMDB movie reviews dataset [33]. In alignment with [28], we utilize the available validation set for each dataset as the test set since test data is not available. However, unlike their setting, which randomly selects fixed number of training samples per dataset (i.e., potentially up-sampling or down-sampling), we employ the original full training set for each dataset to better align with real-world scenarios where the data quantity distribution across tasks is inherently imbalanced. In continual finetuning, we train each task until the training loss converges without validation set. We proceed to train the next task after the previous one is finish, and the training data for each task was no longer available once used.

TABLE 1: The statistic of the 15 downstream tasks.

Task Type	Datasets	# of Train	# of Test
SC	yelp	5000	7600
SC	amazon	5000	7600
NLI	MNLI	3000	7600
NLI	CB	250	56
COPA	COPA	400	100
QQP	QQP	2000	7600
NLI	RTE	2000	277
SC	IMDB	2000	7600
SC	SST-2	2000	872
TC	dbpedia	14000	7600
TC	agnews	4000	7600
SC	yahoo	10000	7600
MultiRC	MultiRC	2000	4848
BoolQA	BoolQA	2000	3270
WiC	WiC	2000	638

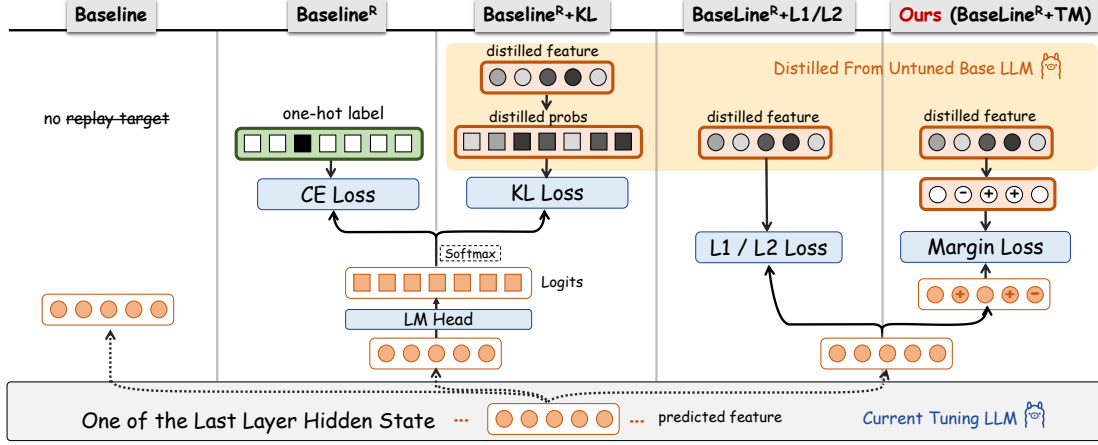


Fig. 3: A comparable baseline series of distinct replay-based optimization targets (left to right): native non-replay Baseline, vanilla replay Baseline^R, replay with different distillation strategies regarding logits imitation Baseline^R+KL and feature imitation Baseline^R+L1/L2. The rightmost Baseline^R+TM is our proposed method, which employs the TM loss.

Table 1 presents the dataset statistics for the 15 tasks. Examples mainly including instructions, inputs, and golden answers for each dataset are provided in the Appendix.

4.2 Metrics

We evaluate the performance of the final model (i.e., after continual finetuning on 15 tasks) from two dimensions comprising general capabilities and the average accuracy over all downstream tasks.

For general capabilities, We employ MMLU [34] benchmark, which spans 57 diverse disciplines ranging from STEM, humanities and social sciences, etc., to rigorously measure both factual knowledge and analytical skills across multiple levels of complexity. We use five-shot setting and discriminative evaluation.

For ability to effectively learn the sequential downstream tasks, we assess the Average Performance (AP) [35] of the final model via obtaining task-wise accuracies and then computing their mean. AP reflects the model’s overall performance across multiple tasks and its ability to retain knowledge from previously learned. Notably, we also evaluated the multi-task learning (MTL) regime which finetunes on the combined dataset of all 15 tasks, serving as the theoretical upper bound performance for continual learning.

Finally, we compute the F1 average of the MMLU and AP to reflect the holistic performance of model in maintaining its original general capabilities while effectively learning downstream tasks. All experimental results are reported as the average of 3 runs.

4.3 Comparable Methods

We meticulously implement all comparable methods from scratch for controlled experiments, including the most basic level and its progressively enhanced counterparts. As shown in Fig. 3, we compare our method (denoted as Baseline^R+TM hereafter) with: native non-replay Baseline, vanilla replay Baseline^R, replay with different distillation strategies regarding logits imitation Baseline^R+KL and feature imitation Baseline^R+L1/L2. These competitors cover the most prevalent and established practices in real-world

application. For fair comparison, all methods are implemented within the same framework using identical replay samples and maintaining consistent configuration throughout the evaluation process. Details are as follows:

- **Baseline:** continually finetune the LLM on sequential tasks without adding any general replay samples.
- **Baseline^R:** continually finetune the LLM on sequential tasks by mixing 1K general replay samples from $\mathcal{D}^{(g)}$ with each task. These samples are pre-selected before finetuning and remain unchanged throughout the entire finetuning process. Both the replay and downstream task samples are jointly optimized using the standard cross-entropy loss. Notably, all the methods discussed subsequently maintain this cross-entropy loss.
- **Baseline^R+KL:** extend the Baseline^R by integrating an additional KL loss. Specifically, the pre-distilled general replay sample logits serve as the target for KL loss during finetuning. The softmax temperature is set to 2, and the weight of KL loss term is accordingly set to 4 (its square) to compensate for gradient scaling down induced by the temperature [36]. In implementation, to avoid the large overhead of pre-storing the high-dimensional *logits* vectors, we compute the *logits* in real-time during finetuning based on the previously acquired final layer hidden state $h^{(g)}$ and the original *lm_head* parameters of LLM.
- **Baseline^R+L1:** extend the Baseline^R by integrating an additional L1 loss computed on the hidden states at the last layer. The previously acquired $h^{(g)}$ serves as the target.
- **Baseline^R+L2:** resemble Baseline^R+L1 but employ L2 loss instead of L1 loss.

Our method and other options are explained as follows:

- **Our method (aka. Baseline^R+TM):** extend the Baseline^R by integrating our proposed TM loss computed on the hidden states at the last layer as in Eq. 6~Eq. 9.
- **BI Option:** adopt Batch Insertion (Sec. 3.2.1) and evaluate its effectiveness across all the replay-based Baseline^R series, i.e., Baseline^R and Baseline^R+KL/L1/L2/TM that typically using general replay samples.
- **Loss Weight:** varied weighting values (denoted as $w=[\]$) of the additional loss term regarding L1/L2/TM are

tested. We first empirically set $w=1$ (omitted as default), $w=100$ and a dynamic weighting $w=d.$ (Sec. 3.2.3) for $\mathcal{L}^{TM}$ to find the optimal performance, and then deliberately evaluate the same optimal weight on L1 and L2 for fair comparison.

- **Upper Bound:** we also include the upper bound performance for comparison, where **Orig** denotes the original MMLU score of untuned base model as ceiling. **MTL** denotes a multi-tasks learning result across all 15 downstream tasks, which is trained on the combined task samples with the identical settings (epochs, learning rate, etc.). We calculate their F1 average upper bound as well.

Notably, $\text{Baseline}^R + \text{L1/L2}$ can be viewed as a stricter version of ours, which pursues precise value fitting (also bringing activation state alignment), but lacks the inherent variation tolerant afforded by our discrete states.

Regarding external competitors, since we strive for a simply yet effective replay-base approach (e.g., prompts-agnostic, task_ids-agnostic, non-generative), we do not compare with ineligible methods like ProgPrompt [28], which sequentially integrates previously learned prompts with the current one during both training and testing. Instead, we compare **O-LoRA** [37] in our LoRA setting due to its simplicity in only constraining the LoRA’s update direction by an additional orthogonal loss term. We are interested in comparing the downstream tasks performance enhanced as a byproduct by our method with that of the specialized O-LoRA, which is solely dedicated to this purpose. We reimplement it carefully using the same LoRA hyperparameters.

TABLE 2: Comparison of different methods on continual full-parameter finetuning (15 epochs per task) in 15 downstream tasks.

Methods (Full-Parameter)	MMLU Score (Final)	15 Tasks AP (Final)	F1 Avg
Baseline	38.3213	37.4720	37.8919
Baseline ^R	50.5332	39.2741	44.1979
w/ BI	55.5556	43.9903	49.1011
Baseline ^R +KL	51.0492	42.0231	46.0985
w/ BI	52.7692	35.5259	42.4638
Baseline ^R +L1	54.9364	66.8605	60.3147
w/ BI	54.5942	66.7673	60.0691
Baseline ^R +L2	55.0052	67.4899	60.6113
w/ BI	56.6219	66.7462	61.2686
Baseline ^R +L1 (w=100)	57.9635	72.4376	64.3973
w/ BI	59.0299	71.1125	64.5103
Baseline ^R +L2 (w=100)	60.7499	72.6112	66.1531
w/ BI	57.8947	73.2265	64.6643
Baseline ^R +L1 (w=d.)	53.1132	67.4546	59.4309
w/ BI	53.2852	64.4925	58.3556
Baseline ^R +L2 (w=d.)	55.1772	71.0094	62.1001
w/ BI	54.7988	68.2590	60.7927
Ours			
Baseline ^R +TM	55.3836	70.3490	61.9756
w/ BI	57.6539	68.7473	62.7138
Baseline ^R +TM (w=100)	60.7155	74.0817	66.7359
w/ BI	60.9907	72.4771	66.2396
Baseline ^R +TM (w=d.)	60.7843	74.4796	66.9386
w/ BI	57.2411	70.5607	63.2068
Upper Bound	Orig 66.5291	MTL 81.0079	73.0580

4.4 Implementation Details

In all experimental comparison, we assess both full-parameter and LoRA finetuning settings with consistent batch size of 64. All downstream samples are truncated with a maximum source length of 512 and a maximum target length of 50. Accordingly general replay samples are truncated with a maximum length of summation 562. In full-parameter setting, we maintain a uniform learning rate of $3e-6$ across all methods, employing a warmup strategy coupled with a cosine learning rate schedule. In LoRA setting, we maintain a uniform learning rate of $1e-4$ with warmup strategy across all methods, while setting LoRA hyperparameters to $r=8$, $\alpha=32$, $\text{LoRA_dropout}=0.1$ and only tuning the parameters limited to q_proj and k_proj . We aim to finetune each task with sufficient steps to ensure loss convergence. Based on preliminary experiments, we ultimately selected 15 epochs per task for the full-parameter setting (due to the smaller learning rate) and 8 epochs for the LoRA.

Notably, the replay samples used in all experiments are identical (the pre-selected set of 1K samples mentioned in Baseline^R), which guarantees that the observed effects are attributable to the methodological variations rather than differences in the replay data. All experiments are conducted using Transformers [38] library with DeepSpeed ZeRO-2 [39] and AdamW optimizer [40], running on up to 8 H800-80GB GPUs.

For the BI option experiments, we set $\rho^{BI}=4/64$, which indicates that 4 general replay samples are inserted into each batch of 64 data points. Specifically, these 4 general replay samples are selected randomly and non-repetitively from the aforementioned 1K samples. Once all samples have been selected, the process is reset to ensure continuously sampling.

4.5 Results

4.5.1 Continual Full-Parameter Finetune

Table 2 shows the performance of each method in continual full finetuning 15 downstream tasks. We find that simply mixing general replay samples (Baseline^R) significantly outperforms the Baseline without any replay, achieving a notable improvement of 12% on MMLU. It justify the widespread adoption of this vanilla replay way in practice.

After additional distillation technique, $\text{Baseline}^R + \text{KL}$ yields further improvements by capturing more information, specifically the distribution of labels. However, under the similar distillation cost, feature-based methods ($\text{Baseline}^R + \text{L1/L2/TM}$) perform remarkable better, empirically suggesting that feature information is more efficient than label information by encoding richer representations of model knowledge in feature layer. Our rationale is that softmax-normalized labels tend to be dominated by extreme values, whereas features preserve finer-grained details. Among them, $\text{Baseline}^R + \text{TM}$ further alleviate the overly rigid inference of L1/L2 loss by appropriately constraining optimization from an activation state perspective, achieving the highest performance.

Moreover, the results of all Baseline^R series validate the hypothesis that using only general replay samples can simultaneously maintain general capabilities and enhance

TABLE 3: Comparison of different methods on continual LoRA finetuning (8 epochs per task) in 15 downstream tasks (w=d. denotes weight dynamic).

Methods (LoRA)	MMLU Score (Final)	15 Tasks AP (Final)	F1 Avg
Baseline	55.7620	73.3944	63.3746
Baseline ^R	58.6515	75.5310	66.0296
w/ BI	56.5187	73.3986	63.8621
Baseline ^R +KL	61.0251	74.8367	67.2289
w/ BI	61.5755	72.9626	66.7872
Baseline ^R +L1	61.5411	73.4170	66.9565
w/ BI	65.1875	73.1178	68.9253
Baseline ^R +L2	61.6787	74.9397	67.6656
w/ BI	64.4651	74.1000	68.9476
Ours			
Baseline ^R +TM	65.3251	75.0639	69.8567
w/ BI	64.6371	72.7650	68.4606
Baseline ^R +TM (w=100)	65.9443	63.9167	64.9147
w/ BI	65.5659	68.7755	67.1323
Baseline ^R +TM (w=d.)	66.2539	64.4417	65.3352
w/ BI	65.4627	67.7580	66.5906
O-LoRA [37]	55.8996	73.6823	63.5707
Upper Bound	Orig 66.5291	MTL 80.3474	72.7882

overall performance of downstream task. As shown in the table, both MMLU and AP improve. This provides an alternative to the traditional practice of laboriously collecting downstream task replay samples during continual finetuning, sparking a promising research direction.

4.5.2 Continual LoRA Finetune

Table 3 shows the performance of each method in continual LoRA finetuning 15 downstream tasks. Compared to full-parameter setting, LoRA alone exhibits notable superiority, which tunes only 0.042% of the parameters (i.e., q_proj and k_proj). This minimal parameter tuning likely contributes to its strong anti-forgetting ability, but it still enables adequate learning of new tasks. For instance, when equipped with LoRA, both the Baseline and vanilla replay Baseline^R nearly match the best F1 Avg observed in full-parameter, and Baseline^R also show surprisingly strong AP of downstream tasks. Still, similar trends hold for LoRA, with the Baseline^R series showing progressive improvements, where our method ultimately achieves the best F1 Avg. We also find that several methods here have achieved MMLU scores nearly matching the upper bound evaluated from original base model, showing negligible loss of general capabilities.

O-LoRA, as a simple and comparable approach dedicated to AP of downstream tasks, achieves decent AP performance but exhibits obvious forgetting of general capabilities. Furthermore, the original O-LoRA paper claims its superiority over the method with replaying downstream task samples, but we still attain a higher AP. This demonstrates the multifaceted advantages of our GeRe framework over tradition.

By the way, the MTL performance under both settings shows that LoRA still slightly underperforms full-parameter when jointly learning multiple new tasks, aligning with study [41] and suggesting that full-parameter remains

preferable in normal situation with available computational resources.

4.5.3 Ablation Study of BI and Loss Weight

Each method with BI option under full-parameter and LoRA settings is additional list in Table 2~3. The effectiveness of BI varies across methods and settings, without showing consistent enhancement. For instance, BI improves the Baseline in full-parameters and Baseline^R+L1/L2 in LoRA, but its effect appears negligible in most distillation methods that already capture more information. We attribute this to the small scale of our finetuning datasets in the experiments, where mixing 1K general replay samples suffices for many downstream tasks. In extremely small datasets like CB, BI’s proportional insertion may even reduce the final replay samples below the standard 1K.

However, BI remain necessary in potential scenarios especially finetuning large-scale downstream task datasets that may far exceeds the 1K replay samples. In such cases, data balancing is crucial since simply mixing them at their original scale leads to insufficient replay. As shown in the results, while BI does not significantly improve performance, it also does not degrade it especially with our method, indicating that Baseline^R+TM with BI can be directly used in most circumstances. In conclusion, the adoption of BI should be determined by practical considerations and an optimal replay insertion ratio, which warrants further investigation.

Regarding different loss weight, the results also vary. In full-parameter setting, our method with dynamic weighting $\mathcal{L}^{TM}$, i.e., Baseline^R+TM (w=d.) performs best, even in a fair comparison where L1 and L2 are purposefully evaluated with the same weighting strategy. In contrast, a simply setting of fixed weight of 1 yields better performance under LoRA, as larger or dynamic weight tend to degrade AP. So We only list results (w=1) of L1 and L2 as well. This indicates that different settings should better have their individual weight strategies, and we have already explore two best practice. Unified settings remain for future research. In the following comparison, we intentionally use results of w=100 for full-parameter setting and w=1 for LoRA in order to simultaneously consider the maximum achievable performance of the L1/L2 competitors.

Furthermore, conventional belief suggests that strengthening the weight of optimization direction toward anti-forgetting will enhance stability at the cost of plasticity, thereby impairing learning of new task. However, results with higher weight (from w=1 to w=100) show that not only MMLU dose but also downstream task’s AP continues to improve. This may stem from the adoption of general replay samples rather than task-specific replay samples, which mitigates the Stability-Plasticity Dilemma [42] typically occurs when excessively replaying samples from downstream tasks in tradition.

4.5.4 Performance Trend Over Tasks

Fig. 4 shows the dynamic changes of the three metrics assessed in Table 2~3 as the model sequentially learns 15 downstream tasks under full-parameter and LoRA settings. Evidently, our method consistently achieves the highest score on MMLU at nearly every task step under both settings. Moreover, it remains in the top tier performance



Fig. 4: Performance trend during continual learning 15 tasks of different methods. Y-axis of each figure indicates the specific task that has just been learned. Two rows depict full-parameter and LoRA settings, respectively. Three columns are metrics: current MMLU score, average performance over tasks learned so far, F1 average.

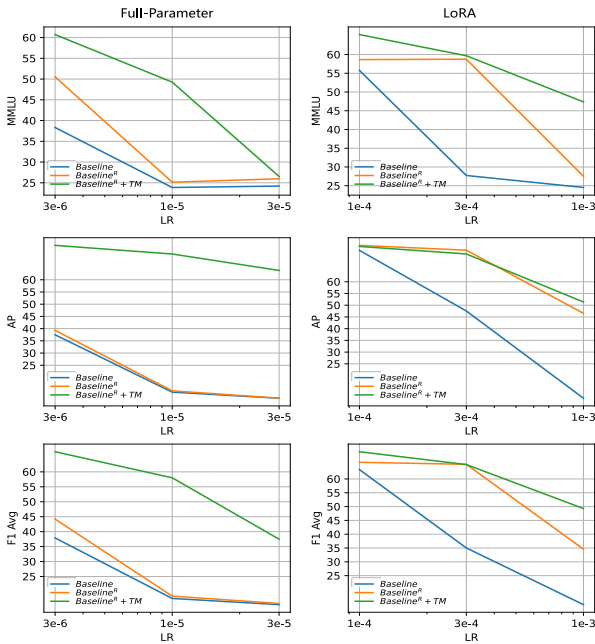


Fig. 5: MMLU, AP and F1 Avg performance of three major representative methods across different learning rate is compared under full-parameter and LoRA settings, with the LR axis displayed on a logarithmic scale.

of AP across the downstream tasks learned so far, achieving the best final F1 Avg.

Notably, during the full-parameter learning, a significant decline in AP occurs after the model learned the COPA task. A Deeper investigation of the task-wise results (see Appendix) reveals that it is mainly caused by performance drops in the MNLI and CB tasks. We attribute this to the unique instruction format of COPA without providing options (see Appendix), which temporarily disrupts the model’s instruction-following ability after learning COPA. So, tasks with the most similar instructions like MNLI and CB experience performance degradation. Fortunately, as the model continues to learn subsequent tasks with regular instructions, the performance of these two tasks recovers. We interpret this as a case of spurious forgetting [43], where the model does not lose the core knowledge of these tasks but undergoes temporary confusion in instruction following, which can be readily restored in later learning phases.

4.5.5 Robustness to Learning Rate

In continual finetuning, it is well-established that while larger learning rates (LR) facilitate more thorough learning of downstream tasks, they also intensify the forgetting of previously acquired knowledge. This effect becomes particularly pronounced when dealing with LLMs featuring

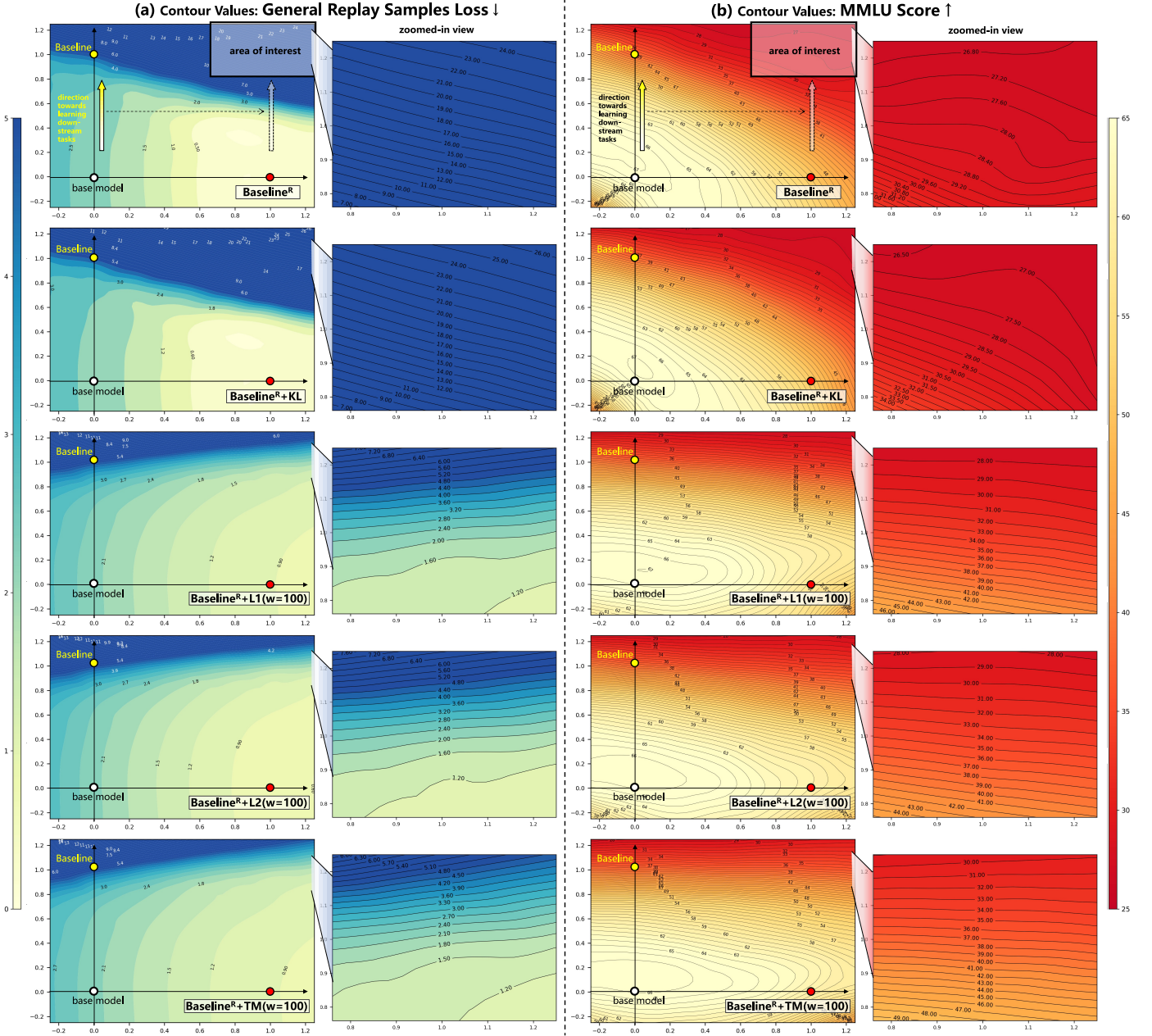


Fig. 6: Landscapes of (a) replay samples loss, and (b) MMLU score under **full-parameter setting**. Origin point (0,0) is base untuned model. Y-axis is weight update direction of Baseline (0,1), representing the learning dedicated to downstream tasks. X-axis is weight update direction of target method for comparison (1,0). The upper-right area of interest simulates the target model guided by the learning direction of downstream tasks (yellow arrow), where the flatness (see zoomed-in view) can imply the optimizing robustness against latent forgetting even under potential overtraining in practice.

massive training parameters, which aligns with our observations in preliminary experiments. Thus, practitioners need to carefully adjust the LR from relatively small values to balance new task acquisition with knowledge retention.

However, empirical evaluating against both the native Baseline and vanilla replay Baseline^R show that our method maintains relatively strong general capabilities (MMLU scores) even with substantially increased LR. As shown in Fig.5, our method gains more stable performance despite a 3× LR increase under full-parameter and a 10× increase under LoRA. In contrast, the compared methods approach a MMLU score of nearly 25%, equivalent to random guessing

among 4 options, highlighting their vulnerable dependence on tuning LR. Our methods demonstrates superior adaptability in practice scenarios. Beyond MMLU, similar conclusions regarding AP and the resulting F1 Avg can be drawn from the subsequent subfigures.

4.5.6 Robustness in Optimization Landscape

To better understand the underlying optimization mechanisms of different methods and their robustness against forgetting, we visualize the landscapes with two contour values under full-parameter (Fig.6) and LoRA settings (Fig.7)

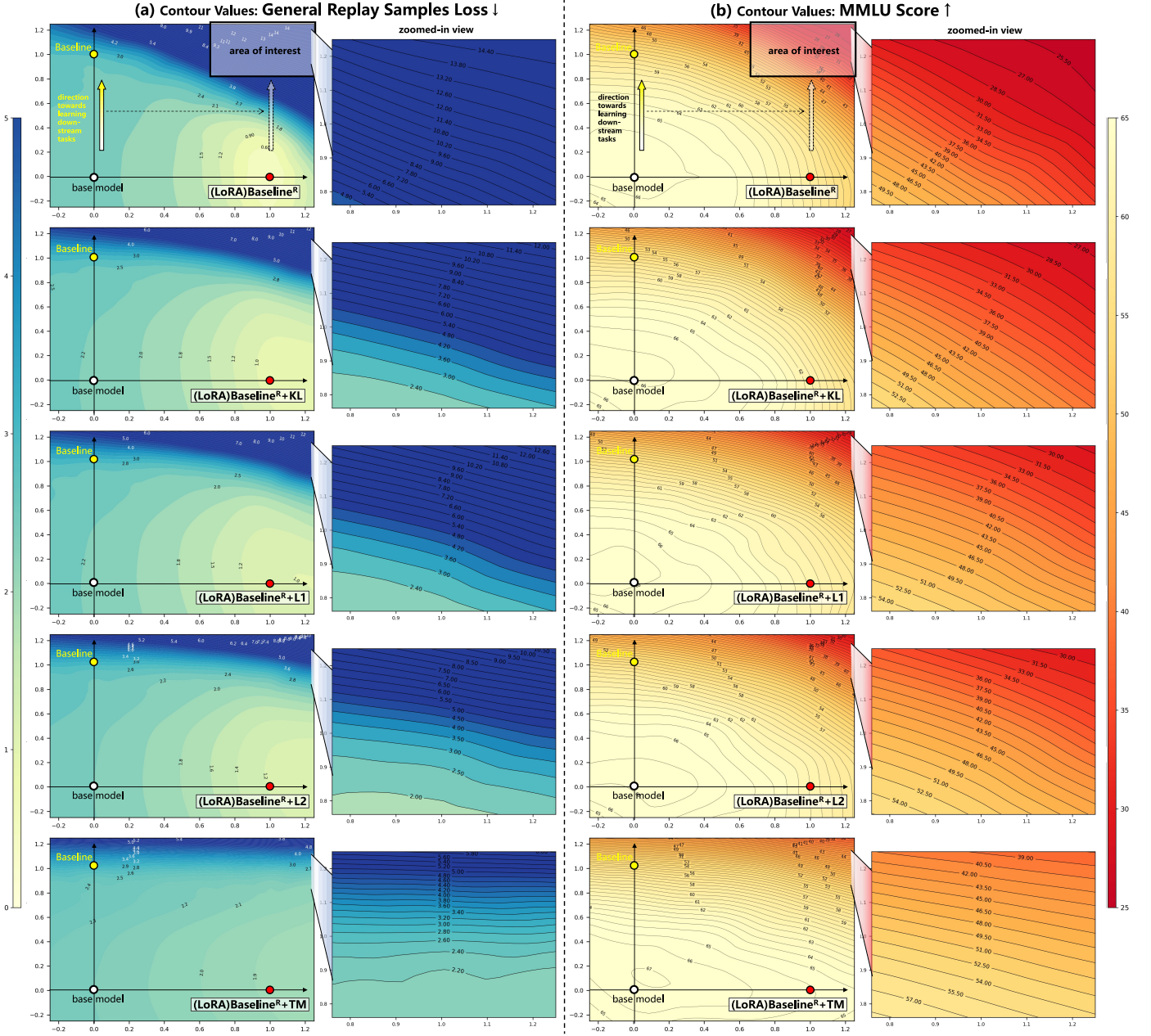


Fig. 7: Landscapes of (a) replay samples loss, and (b) MMLU score under **LoRA setting**. Origin point (0,0) is base untuned model. Y-axis is weight update direction of Baseline (0,1), representing the learning dedicated to downstream tasks. X-axis is weight update direction of target method for comparison (1,0). The upper-right area of interest simulates the target model guided by the learning direction of downstream tasks (yellow arrow), where the flatness (see zoomed-in view) can imply the optimizing robustness against latent forgetting even under potential overtraining in practice.

Our idea is that in replay-based learning, to ensure thorough learning for downstream tasks, excessive training can easily occur, which will compromise the general capabilities. We consider that a better method should reconcile the optimization directions of both downstream tasks and replay samples. Such a method would maintain latent robustness even when subjected to excessive downstream task training that typically induces forgetting. According to task vector arithmetic [44], we can directly perform linear combinations of model weight for a specific method to simulate and observe their robustness to optimization dynamics under extreme conditions of excessive training. Therefore, the

landscape is designed as a 2D weight space spanned by two specific model weight update directions, where y-axis is the update direction of the Baseline and x-axis is one of the interested methods for comparison. Specifically, the upward direction (yellow arrow) indicates the optimization toward native continual finetuning without any replay, highlighting exclusive learning of downstream tasks. The rightward direction indicates the optimization toward a target model trained from a specific method among the replay-based series. Based on this coordinate, the upper-right region (white rectangle) is the area of interest that can reveal the robustness against forgetting undergoing potential overtraining,

as it simulates the weight update direction imposed on the target model toward overly optimizing downstream tasks.

As for contour values, we select two metrics: a) the CE loss of replay samples, as it is universally adopted and optimized across all methods, and b) the direct MMLU score for straightforward observation. These values measure the retention of general capabilities implicitly and explicitly, respectively. The flatness (i.e., the rate of performance degradation) of the area of interest indicates how robustly each method preserves their general capabilities while learning downstream task.

The landscapes are implemented with positioning the untuned base LLM model at coordinate (0,0), the native Baseline finetuned model at coordinate (1,0), and a specific replay-based finetuned model for comparison at coordinate (0,1). Weight parameters of each model are flattened into a vector, and the two basis vectors of this coordinate system are derived by subtracting the corresponding model weight vectors (e.g., $\vec{y} = \mathbf{w}_{[\text{target}]} - \mathbf{w}_{\text{base}}$, $\vec{x} = \mathbf{w}_{\text{baseline}} - \mathbf{w}_{\text{base}}$). We use these two basis vectors to generate a grid of points, where each point represents a model derived from a linear combination of these basis vectors (e.g., (0.6,0.4) denotes model of weight $\mathbf{w} = 0.6\vec{x} + 0.4\vec{y}$). For every model associated with the points, we compute the loss of replay samples and the MMLU score, creating the contour plot as landscape, respectively.

For instance, Fig.6(a) shows the replay sample loss under full-parameter setting. We observe that the Baseline^R exhibits a notably steepness in the area of interest, implying that this optimization encounters significant conflicts when following the direction of learning downstream tasks while attempting to replay to retain general capabilities. This potentially accounts for its poor performance in Tab.2. The same issue persists in Baseline^R+KL, though it is relatively less severe, but still remarkable. In comparison, the feature-based replay methods show significantly flatter behavior in the area of interest, and among them our Baseline^R+TM performs the flattest upon closer look at the contour values in the zoomed-in view. This mean our method show better robustness to the intrinsic optimization dynamics when arbitrarily or excessively trained on downstream tasks.

For the MMLU score landscape in Fig.6(b), similar trend is observed. The feature-based replay methods exhibit higher scores and slower decline in the area of interest. They also shape a distinct ridge along the learning trajectory (i.e., y-axis) of the specific model, where the scores are maximally preserved. Besides, though the learning trajectories of baseline^R and baseline^R+KL maintain a high score early, they undergo sharply decline as more tasks are introduced. In contrast, the feature-based methods effectively preserve the score along the ridge, confirming the necessity of benchmarking typical long-sequence tasks. In the MMLU score landscape, our Baseline^R+TM still demonstrate superior performance.

Additionally, although the contour patterns of general samples loss and MMLU score differ, their underlying trends exhibit similar characteristics, e.g., both metrics show consistent variations in the area of interest of the same method. This confirms that general samples can implicitly reflect the actual general capabilities. However, relying solely on the CE loss of general samples may be insufficient,

e.g., Baseline^R and Baseline^R+KL achieve lower loss values, but their MMLU scores remain low. Instead, the optimization of feature-based methods align more closely with the trends of MMLU.

Finally, Fig.7 shows landscape under LoRA setting, where the observations are generally similar to those of full-parameter, except that LoRA—as a highly effective anti-forgetting tool—significantly enhances the foundational performance of all variants. Notably, our method here shows more pronounced flatness and maintains higher performance in the are of interest. Across both settings, our Baseline^R+TM consistently demonstrates the optimizing robustness against latent forgetting.

5 CONCLUSION

In this research, we introduce GeRe, a framework that leverages general replay samples for continual learning in LLMs. Building upon GeRe, we revisit the existing replay baseline and devise a novel optimization method that utilizes the informative states of neurons through a proposed TM loss. This loss function effectively aligns the activation states of replay samples, offering a moderate yet discerning constraint compared to existing replay-based variants. Crucially, GeRe’s results reveal that only a fixed set of general replay samples is sufficient for continual learning across a long sequence of downstream tasks, which not only effectively retains the general capabilities but also successfully promotes the overall performance on downstream tasks. Furthermore, detailed analyses and intuitive visualizations rigorously validate the superior performance and robustness of the TM loss within GeRe. Our study offers valuable insights into the efficacy of replay mechanisms, highlighting the practical advantages and contributing to potential applications for the continuous iteration of LLMs.

REFERENCES

- [1] Y. Luo, Z. Yang, F. Meng, Y. Li, J. Zhou, and Y. Zhang, “An empirical study of catastrophic forgetting in large language models during continual fine-tuning,” *arXiv preprint arXiv:2308.08747*, 2023. 1
- [2] J. Zheng, S. Qiu, and Q. Ma, “Concept-1k: A novel benchmark for instance incremental learning,” *arXiv e-prints*, pp. arXiv–2402, 2024. 1
- [3] H. Shi, Z. Xu, H. Wang, W. Qin, W. Wang, Y. Wang, Z. Wang, S. Ebrahimi, and H. Wang, “Continual learning of large language models: A comprehensive survey,” *arXiv preprint arXiv:2404.16789*, 2024. 1
- [4] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale *et al.*, “Llama 2: Open foundation and fine-tuned chat models,” *arXiv preprint arXiv:2307.09288*, 2023. 2
- [5] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei *et al.*, “Qwen2. 5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024. 2
- [6] B. A. Olshausen and D. J. Field, “Emergence of simple-cell receptive field properties by learning a sparse code for natural images,” *Nature*, vol. 381, no. 6583, pp. 607–609, 1996. 2
- [7] J. Wolfe, A. R. Houweling, and M. Brecht, “Sparse and powerful cortical spikes,” *Current opinion in neurobiology*, vol. 20, no. 3, pp. 306–312, 2010. 2
- [8] G. I. Parisi, R. Kemker, J. L. Part, C. Kanan, and S. Wermter, “Continual lifelong learning with neural networks: A review,” *Neural networks*, vol. 113, pp. 54–71, 2019. 2
- [9] R. M. French, “Catastrophic forgetting in connectionist networks,” *Trends in cognitive sciences*, vol. 3, no. 4, pp. 128–135, 1999. 2

- [10] D. Rolnick, A. Ahuja, J. Schwarz, T. Lillicrap, and G. Wayne, "Experience replay for continual learning," *Advances in neural information processing systems*, vol. 32, 2019. 2
- [11] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the national academy of sciences*, vol. 114, no. 13, pp. 3521–3526, 2017. 2
- [12] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE transactions on pattern analysis and machine intelligence*, vol. 40, no. 12, pp. 2935–2947, 2017. 3
- [13] A. A. Rusu, N. C. Rabinowitz, G. Desjardins, H. Soyer, J. Kirkpatrick, K. Kavukcuoglu, R. Pascanu, and R. Hadsell, "Progressive neural networks," *arXiv preprint arXiv:1606.04671*, 2016. 3
- [14] A. Mallya, D. Davis, and S. Lazebnik, "Piggyback: Adapting a single network to multiple tasks by learning to mask weights," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 67–82. 3
- [15] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen *et al.*, "Lora: Low-rank adaptation of large language models," *ICLR*, vol. 1, no. 2, p. 3, 2022. 3
- [16] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015. 3
- [17] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter," *arXiv preprint arXiv:1910.01108*, 2019. 3
- [18] X. Jiao, Y. Yin, L. Shang, X. Jiang, H. Chen, L. Li, F. Wang, and Q. Liu, "Tinybert: Distilling bert for natural language understanding," *arXiv preprint arXiv:1909.10351*, 2020. 3
- [19] Y. Zhou, Y. Wang, J. Zhang, and X. Li, "Distilling task-specific knowledge from large pre-trained models," *arXiv preprint arXiv:2203.12345*, 2022. 3
- [20] Z. Zhengetal, "Localizationdistillationforobjectdetection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 10070–10083, 2023. 3
- [21] L. Wang, M. Zhang, and Y. Liu, "Self-distillation for large language models," *arXiv preprint arXiv:2301.04567*, 2023. 3
- [22] X. Xu, M. Li, C. Tao, T. Shen, R. Cheng, J. Li, C. Xu, D. Tao, and T. Zhou, "A survey on knowledge distillation of large language models," *arXiv preprint arXiv:2402.13116*, 2024. 3
- [23] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan *et al.*, "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024. 5
- [24] H. Li, Z. Xu, G. Taylor, C. Studer, and T. Goldstein, "Visualizing the loss landscape of neural nets," *Advances in neural information processing systems*, vol. 31, 2018. 5
- [25] L. Wang, X. Zhang, H. Su, and J. Zhu, "A comprehensive survey of continual learning: Theory, method and application," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2024. 5
- [26] D. Soboleva, F. Al-Khateeb, R. Myers, J. R. Steeves, J. Hestness, and N. Dey, "Slimpajama: A 627b token cleaned and deduplicated version of redpajama," 2023. [Online]. Available: <https://huggingface.co/datasets/cerebras/SlimPajama-627B> 5
- [27] M. Weber, D. Y. Fu, Q. Anthony, Y. Oren, S. Adams, A. Alexandrov, X. Lyu, H. Nguyen, X. Yao, V. Adams, B. Athiwaratkun, R. Chalamala, K. Chen, M. Ryabinin, T. Dao, P. Liang, C. Ré, I. Rish, and C. Zhang, "Redpajama: an open dataset for training large language models," *NeurIPS Datasets and Benchmarks Track*, 2023. 5
- [28] A. Razdaibiedina, Y. Mao, R. Hou, M. Khabsa, M. Lewis, and A. Almahairi, "Progressive prompts: Continual learning for language models," in *The Eleventh International Conference on Learning Representations*, 2023. 5, 7
- [29] X. Zhang, J. Zhao, and Y. LeCun, "Character-level convolutional networks for text classification," *Advances in neural information processing systems*, vol. 28, 2015. 5
- [30] C. Qin and S. Joty, "Lfpt5: A unified framework for lifelong few-shot language learning based on prompt tuning of t5," *arXiv preprint arXiv:2110.07298*, 2021. 5
- [31] A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman, "Glue: A multi-task benchmark and analysis platform for natural language understanding," *arXiv preprint arXiv:1804.07461*, 2018. 5
- [32] A. Wang, Y. Pruksachatkun, N. Nangia, A. Singh, J. Michael, F. Hill, O. Levy, and S. Bowman, "Superglue: A stickier benchmark for general-purpose language understanding systems," *Advances in neural information processing systems*, vol. 32, 2019. 5
- [33] A. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning word vectors for sentiment analysis," in *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*, 2011, pp. 142–150. 5
- [34] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," *arXiv preprint arXiv:2009.03300*, 2020. 6
- [35] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 532–547. 6
- [36] S. Sun, W. Ren, J. Li, R. Wang, and X. Cao, "Logit standardization in knowledge distillation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 15731–15740. 6
- [37] X. Wang, T. Chen, Q. Ge, H. Xia, R. Bao, R. Zheng, Q. Zhang, T. Gui, and X. Huang, "Orthogonal subspace learning for language model continual learning," *arXiv preprint arXiv:2310.14152*, 2023. 7, 8
- [38] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz *et al.*, "Transformers: State-of-the-art natural language processing," in *Proceedings of the 2020 conference on empirical methods in natural language processing: system demonstrations*, 2020, pp. 38–45. 7
- [39] S. Rajbhandari, J. Rasley, O. Ruwase, and Y. He, "Zero: Memory optimizations toward training trillion parameter models," in *SC20: International Conference for High Performance Computing, Networking, Storage and Analysis*. IEEE, 2020, pp. 1–16. 7
- [40] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint arXiv:1711.05101*, 2017. 7
- [41] D. Biderman, J. Portes, J. J. G. Ortiz, M. Paul, P. Greengard, C. Jennings, D. King, S. Havens, V. Chiley, J. Frankle *et al.*, "Lora learns less and forgets less," *arXiv preprint arXiv:2405.09673*, 2024. 8
- [42] S. Dohare, J. F. Hernandez-Garcia, Q. Lan, P. Rahman, A. R. Mahmood, and R. S. Sutton, "Loss of plasticity in deep continual learning," *Nature*, vol. 632, no. 8026, pp. 768–774, 2024. 8
- [43] J. Zheng, X. Cai, S. Qiu, and Q. Ma, "Spurious forgetting in continual learning of language models," in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=ScL7IiKGdI> 9
- [44] G. Ilharco, M. T. Ribeiro, M. Wortsman, L. Schmidt, H. Hajishirzi, and A. Farhadi, "Editing models with task arithmetic," in *The Eleventh International Conference on Learning Representations*, 2023. [Online]. Available: <https://openreview.net/forum?id=6t0KwF8-jrj> 11

APPENDIX

A

.1 Examples of General Replay Samples

Table 4 presents some selected examples from the general replay samples set $\mathcal{D}^{(g)}$ used throughout this paper, where each entry is a normal pretraining text sentence. The ID is the line number of our released *jsonl* file of $\mathcal{D}^{(g)}$, and the set name is the meta information indicating the source. (We purposely selected examples from diverse sources for display.) The data was obtained through the following process: We downloaded the first chunk of data (*train-00000-of-00048-ab2b35705f029d94.parquet*) from SlimPajama-6B (<https://huggingface.co/datasets/DKYoon/SlimPajama-6B>), a sampled version of SlimPajama-627B (<https://huggingface.co/datasets/cerebras/SlimPajama-627B>). Then, we simply extracted the first 1000 entries using the following code to generate the *jsonl* file:

```
slim_datasets = load_dataset('parquet',
                             data_files={'train-00000-of-00048-
                                           ab2b35705f029d94.parquet'})
slim_datasets.select(range(1000)).to_json('slimpajama_6B_chunk0_head1k.jsonl')
```

Id	Set Name	Text
0	RedPajamaC4	Want Tori to Coach You Too? Tori's Health Step by Step coming soon. Win free copies, prizes, access to exclusive behind-the-scenes, free access to Coach Tori, and more. and receive a copy of Tori's Weekly Challenges. We'll also notify you of when Tori's Program becomes available. I've been asked, even criticized, about adding a focus on nutrition to Desert. There's a reason why. I had poor nutritional examples growing up. Being confused on the issue of nutrition cost me a lot. I remember yo-yo'ing a lot. The only time I even came close to being my desired weight was when I did high-intensity workouts daily. At one point, I was exercised about 6 hours a day. I was in multiple dance classes and a karate class, as well as another karate club that met for two hours three days a week. I also rode my bike to campus, and even added a one hour workout when I got home. I was still thirty pounds overweight. I can attest to the coined phrase "You cannot exercise away a bad diet." It was hard to consider diet for me, because I had a genetic heritage that leaned on the heavy side. I felt trapped, having a low metabolism. It seemed if I even looked at what others ate, I was the one who gained weight. Every once in a while, someone would mention diet to me, but it did little to sway me. Why? Bad examples. Brad Pitt in Ocean's Eleven. Every scene he's in, he's eating something unhealthy. In Hollywood and at school, lots of "lean" people were eating the things I loved: pizza, ice cream, hamburgers, fries, bread, cake, cookies, etc. I also knew several "weighed down" people who were eating a healthy diet. It wasn't until I was in college, having just finished my laps in swimming plus a jog, that I stopped by to visit a friend—a slim friend who never seemed hungry. It seemed so unfair as I watched her prepare herself a salad and two small slices of pizza. I knew in that moment that if I ate like her, I would look like her. I also knew that if I prepared a small salad and two small slices of pizza, that by the end of the meal I would end up not eating just one personal pizza, but two or three. I started to believe in nutrition, but I didn't have faith that someone like me could do it. I was right, and I was wrong . . .
25	RedPajamaGithub	ACCEPTED According to The Catalogue of Life, 3rd January 2011 Published in New Zealand J. Bot. 25:166. 1987 Original name Atropis pumila Kirk Remarks
500	RedPajamaC4	Located in an impressive old draper's warehouse, Citibase Birmingham Mailbox is in the heart of the vibrant Mailbox shopping, entertainment and dining district. The recently refurbished reception area is reminiscent of a New York warehouse and the centre provides a wide range of offices many with amazing city views, including new Loft-style suites. With New Street Station and the smart new Grand Central Shopping Centre, the Central Business District and the vast array of other shopping and dining options all under 10 minutes' walk away, it's the ideal location to grow your business.
839	RedPajama CommonCrawl	Big Boy's 24/7 Channel Real 92.3 LA BigBoyTV Videos Big Boy's Bankroll Big Boy Full Episodes Big Boy's Fully Loaded Interviews Big Boy's Uncut Podcast What's Trending with Natalia Perez Meet the Neighborhood Natalia Perez Vick One DJ Hed Advertise on Big Boy's Neighborhood Podcast: Home Grown Radio Tupac's "Strictly 4 My NIGGAZ" Will Be Out Again For Its 25th Anniversary By DJ Hed Feb 17, 2018 Yesterday (February 16) marked the 25th anniversary of Tupac's sophomore album, Strictly 4 My N.I.G.G.A.Z. Today, Interscope Records and UMe are gifting Pac and hip-hop fans are around the world, the blessing to cop a limited edition, commemorative 2LP vinyl of the project. There's two vinyl editions available for purchase. The standard edition is available at all physical retailers and comes with the 180-gram vinyl of the album, and the deluxe edition features a gatefold image of 2PAC's original notebook, with his handwritten track list visible, and prints. The only catch is that there's only 1,000 copies made, which means you have to cop your copy asap. Grab your Strictly 4 My N.I.G.G.A.Z copy on Tupac's website here. About DJ Hed DJ Hed is a deejay mixer on REAL 92.3 KRRL FM Los Angeles Radio Read More Big Boy Blog Big Boy's Full Episodes BIGBOY Political File u00a9 2021 Premiere Networks, Inc.
1000	RedPajamaC4	The Pastel Piebald is a co-dom recessive morph combination, we produced it in 2005 along with The Snake Keeper. After missing the odds on multiple clutches, our luck changed with the second to last clutch of the season, from a 5 egg clutch of Pastel het Pied x het Pied, out came one of our prized possessions one of the first Pastel Pieds. You can imagine the excitement and joy that was felt on that Labor Day holiday in 2005, when we discovered this beautiful Pastel Pied had hatched and it was a male. This male has grown up and in 2008 had sired the first Super Pastel Pied or "Killer Pied", a stunning lemon yellow Piebald, creating a greater demand for the already sought after Pastel Pied.

TABLE 4: The example of the adopted general replay samples

This acquisition process demonstrates that the general replay sample set was obtained through random selection rather than deliberate curation, thereby substantiating the robustness and universality of our method regarding the replay data.

.2 Examples of Downstream Task Datasets

Table.5~6 show detailed examples of 15 downstream task datasets, including task types, dataset names, instructions, inputs, and golden answers. All data are constructed using the same template:

[Instruction]\n[Input]\nAnswer:[Golden Answer].

The evaluation criterion for all samples is binary classification for being correct or incorrect, determining whether the model-generated answers exactly match the golden answers. The accuracy for each dataset is then calculated as the corresponding task performance.

.3 Task-wise Results On Continual Fine-tuning 15 Downstream Tasks

We show the task-wise results of the continual fine-tuning experiments as in Table.7~44, corresponding to each entry in Table.2 and Table.3. For instance, Table.7 elaborates the Baseline entry in Table.2 by including performance of the previous 14 tasks, rather than only listing the final result of the last task. The first column indicates the current learning task, while the remaining columns show evaluation metrics for previously learned tasks and the resulted MMLU and F1 Avg at each learning step.

Task Type	Dataset	Instruction	Input	Golden Answer
SC	yelp	What is the sentiment of the following paragraph? Choose one from the option. Option: very negative, negative, neutral, positive, very positive	Text: This place is printing money and rightfully so. They simply do a bang up job. Best BBQ in AZ.	very positive
SC	amazon	What is the sentiment of the following paragraph? Choose one from the option. Option: very negative, negative, neutral, positive, very positive	Title: Very fragile...arrived broken Text: The set is cute, but refrigerator door was broken on arrival and not repairable. The table top and hutch had come apart and required regluing. This set will not stand up to play.	negative
NLI	MNLI	What is the logical relationship between the "sentence 1" and the "sentence 2"? Choose one from the option. Option: neutral, entailment, contradiction	sentence 1: She leaned back in her chair. sentence 2: She stood next to a chair.	neutral
NLI	CB	What is the logical relationship between the "sentence 1" and the "sentence 2"? Choose one from the option. Option: entailment, contradiction, neutral	sentence 1: A: Your turn. B: Okay. Uh, I don't think they should abolish it. sentence 2: they should abolish it	contradiction
COPA	COPA		Which sentence is the cause of "I coughed."? Choose one between A and B. A: I inhaled smoke. B: I lowered my voice.	A
QQP	QQP	Whether the "first sentence" and the "second sentence" have the same meaning? Choose one from the option. Option: False, True	first sentence: What are the best franchises in India? second sentence: What are the best franchise in India?	True
NLI	RTE	What is the logical relationship between the "sentence 1" and the "sentence 2"? Choose one from the option. Option: contradiction, entailment	sentence 1: The girl was found in Drummondville. sentence 2: Drummondville contains the girl.	contradiction
SC	IMDB	What is the sentiment of the following paragraph? Choose one from the option. Option: Good, Bad	This is a good film. This is very funny. Yet after this film there were no good Ernest films!	Good
SC	SST-2	What is the sentiment of the following paragraph? Choose one from the option. Option: Good, Bad	Text: it 's not the ultimate depression-era gangster movie .	Bad
TC	dbpedia	What is the topic of the following paragraph? Choose one from the option. Option: Company, Educational Institution, Artist, Athlete, Office Holder, Mean of Transportation, Building, Natural Place, Village, Animal, Plant, Album, Film, Written Work	Title: Cori Schumacher Text: Cori Schumacher is a world champion surfer from California.	Athlete
TC	agnews	What is the topic of the following paragraph? Choose one from the option. Option: World, Sports, Business, Science or Technology	Title: British sailors bag bronze Text: Britain's Chris Draper and Simon Hiscocks win bronze in a tense final 49er race on the Saronic Gulf.	Sports
TC	yahoo	What is the topic of the following paragraph? Choose one from the option. Option: Society & Culture, Science & Mathematics, Health, Education & Reference, Computers & Internet, Sports, Business & Finance, Entertainment & Music, Family & Relationships, Politics & Government	Title: did God create people or did people create god?.. Question: think about it.. Answer: Good question dude.	Society & Culture

TABLE 5: The example of the 15 downstream tasks for fine-tuning

MultiRC	MultiRC	According to the following passage and question, is the candidate answer true or false? Choose one from the option. Option: False, True	paragraph: Susan wanted to have a birthday party. She called all of her friends. She has five friends. Her mom said that Susan can invite them all to the party. Her first friend could not go to the party because she was sick. Her second friend was going out of town. Her third friend was not so sure if her parents would let her. The fourth friend said maybe. The fifth friend could go to the party for sure. Susan was a little sad. On the day of the party, all five friends showed up. Each friend had a present for Susan. Susan was happy and sent each friend a thank you card the next week. question: Did Susan call her friends before or after asking her mother? candidate answer: Before asking her mother	True
BoolQA	BoolQA	According to the following passage, is the question true or false? Choose one from the option. Option: True, False	question: can u drive in canada with us license passage: American entry into Canada by land – Persons driving into Canada must have their vehicle’s registration document and proof of insurance.	True
WiC	WiC	Given a word and two sentences, whether the word is used with the same sense in both sentence? Choose one from the option. Option: True, False	word: touch He has a touch of rheumatism. He longed for the touch of her hand.	False

TABLE 6: The example of the 15 downstream tasks for fine-tuning

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	63.8440	62.0571	65.7368	65.7368	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	60.9339	58.1699	63.9737	63.4605	64.4868	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	58.7423	55.0740	62.9342	53.8158	49.8158	85.1711	-	-	-	-	-	-	-	-	-	-	-	-
CB	57.8043	55.3492	60.4874	51.4737	47.4605	82.2500	98.2143	-	-	-	-	-	-	-	-	-	-	-
COPA	31.0482	52.8724	21.9768	36.4605	28.6711	0.0000	0.0000	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	57.5412	53.5604	62.1613	54.8289	55.7500	52.6447	51.7857	98.0000	85.0263	-	-	-	-	-	-	-	-	-
RTE	57.4316	51.8060	64.4277	55.0789	56.0921	61.4211	87.5000	97.0000	83.5921	89.8917	-	-	-	-	-	-	-	-
IMDB	58.2702	51.4964	67.0960	51.7237	54.2237	62.2368	87.5000	96.0000	83.6842	89.8917	82.2500	-	-	-	-	-	-	-
SST-2	55.5752	49.9484	62.6307	45.5921	42.8816	63.3026	89.2857	91.0000	74.2368	89.8917	81.9474	94.2661	-	-	-	-	-	-
dbpedia	52.2713	48.6068	56.5334	30.0263	24.3421	62.8684	85.7143	96.0000	80.7763	74.0072	40.6316	57.5688	99.0658	-	-	-	-	-
agnews	54.6519	46.4740	66.3224	41.2500	30.2237	63.0658	82.1429	42.0000	77.8026	59.9278	71.4474	92.6606	85.5921	92.2895	-	-	-	-
yahoo	41.6585	38.4933	45.3909	30.1184	5.4474	39.9342	50.0000	54.0000	27.7368	40.7942	31.0263	55.5046	77.7368	77.0132	72.9737	-	-	-
MultiRC	53.1073	47.5748	60.0959	41.3816	29.5658	24.1711	19.6429	55.0000	82.4605	64.2599	81.5263	93.8073	90.3421	72.5921	46.1447	74.0924	-	-
BoolQA	51.7833	47.6436	56.7108	42.3289	29.3684	10.0658	8.9286	29.0000	80.1184	17.3285	80.5789	93.8073	90.8553	69.0658	27.9211	72.0297	83.4557	-
WiC	37.8919	38.3213	37.4720	12.2500	4.9605	0.5132	0.0000	12.0000	60.5000	1.0830	39.8684	52.6376	83.8289	55.8816	13.1316	57.5908	68.7462	70.2194

TABLE 7: The task-wise performance (%) of Baseline under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.6890	62.9859	66.4868	66.4868	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	62.8706	61.1971	64.6382	64.6579	64.6184	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	59.6392	56.0028	63.7807	55.9211	50.0395	85.3816	-	-	-	-	-	-	-	-	-	-	-	-
CB	57.6580	53.4228	62.6225	56.3026	50.5132	80.8158	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	46.5305	57.0003	39.3100	51.5395	45.2763	20.5921	8.9286	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	57.5918	55.6244	59.7035	58.6579	54.3816	39.7763	48.2143	97.0000	85.5921	-	-	-	-	-	-	-	-	-
RTE	58.7280	54.3172	63.9185	56.2632	50.3553	62.8158	83.9286	95.0000	84.7368	89.8917	-	-	-	-	-	-	-	-
IMDB	61.4086	55.4868	68.7456	56.0658	55.5526	63.0658	85.7143	95.0000	85.1053	88.8087	82.7368	-	-	-	-	-	-	-
SST-2	59.5391	54.3172	65.8720	51.5132	49.1842	63.2632	85.7143	96.0000	78.2105	89.5307	82.3158	96.1009	-	-	-	-	-	-
dbpedia	57.9254	52.4596	64.6626	47.7237	42.7895	63.6842	85.7143	97.0000	85.1711	88.0866	47.9474	66.9725	98.9605	-	-	-	-	-
agnews	58.9884	50.3612	71.1825	48.3553	44.1711	62.8289	83.9286	38.0000	84.3947	68.2310	76.9737	92.8899	86.6184	92.8947	-	-	-	-
yahoo	44.4589	42.1053	47.0912	26.9737	3.8421	48.4079	60.7143	15.0000	29.4211	26.7148	38.9342	47.7064	72.9737	83.2895	73.8816	-	-	-
MultiRC	52.5617	53.1820	51.9558	37.8026	27.0526	42.3684	37.5000	43.0000	84.7237	56.3177	75.5658	92.6606	67.5526	44.7895	15.8158	76.0520	-	-
BoolQA	53.9872	55.3492	52.6907	41.6316	27.7632	31.3289	32.1429	40.0000	82.5789	46.5704	78.9342	92.6606	68.5789	42.5526	16.6053	74.1337	84.8318	-
WiC	44.1979	50.5332	39.2741	23.9211	14.6579	19.0395	17.8571	27.0000	61.4342	16.6065	67.3553	75.5734	42.8289	32.7237	12.9474	69.8226	71.5596	73.6677

TABLE 8: The task-wise performance (%) of Baseline^R under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.0879	62.4011	65.8684	65.8684	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	62.5326	60.6467	64.5395	64.8158	64.2632	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	57.3472	58.2387	56.4825	46.3421	37.6447	85.4605	-	-	-	-	-	-	-	-	-	-	-	-
CB	53.2286	58.1355	49.0856	39.6974	31.0263	76.2368	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	44.9626	57.7227	36.8226	58.0789	51.6974	0.2105	0.0000	94.0000	-	-	-	-	-	-	-	-	-	-
QQP	61.5757	57.9291	65.7121	63.8158	61.5921	51.4342	80.3571	88.0000	85.6053	-	-	-	-	-	-	-	-	-
RTE	60.0372	54.9364	66.1823	61.6579	57.7763	60.7368	80.3571	85.0000	83.2895	91.3357	-	-	-	-	-	-	-	-
IMDB	60.8189	55.2460	67.6424	54.4211	53.9868	61.7895	83.9286	83.0000	84.2237	90.9747	82.6184	-	-	-	-	-	-	-
SST-2	60.7176	55.9340	66.3961	49.8816	49.4868	62.3158	83.9286	90.0000	83.7895	90.6137	81.8421	95.5275	-	-	-	-	-	-
dbpedia	60.6662	55.2460	67.2657	46.5921	46.6316	64.0000	87.5000	95.0000	84.3421	89.1697	59.6711	83.6009	99.1711	-	-	-	-	-
agnews	62.0695	54.0420	72.8979	48.0526	47.7105	63.8026	85.7143	84.0000	84.2500	88.0866	82.2368	95.5275	88.0658	92.7763	-	-	-	-
yahoo	52.3671	48.4004	57.0421	35.9605	13.1316	52.4868	73.2143	31.0000	57.2368	62.4549	55.3289	84.7477	78.8158	86.7368	73.4868	-	-	-
MultiRC	56.9748	57.1723	56.7786	44.7237	41.2632	40.6711	37.5000	68.0000	83.8947	63.5379	67.8289	91.7431	76.5921	60.2105	22.8947	75.4125	-	-
BoolQA	56.0990	58.3419	54.0222	46.5658	38.4342	12.6447	0.0000	57.0000	80.9079	14.8014	71.6974	86.2385	75.7368	59.6184	18.8553	74.5050	83.6391	-
WiC	49.1011	55.5556	43.9903	30.5263	14.2105	2.5000	1.7857	43.0000	72.5395	3.2491	59.7895	62.2706	66.3816	55.6842	16.6711	69.8845	78.1346	69.2790

TABLE 9: The task-wise performance (%) of Baseline^R with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.8132	63.5707	66.1053	66.1053	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	62.3487	60.4403	64.3816	64.8158	63.9474	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	61.7917	56.2092	68.6053	63.0395	56.9211	85.8553	-	-	-	-	-	-	-	-	-	-	-	-
CB	61.9254	56.4499	68.5772	63.0526	57.6711	84.7763	100.0000	-	-	-	-	-	-	-	-	-	-	-
COPA	45.0773	56.1060	37.6721	55.6184	53.0789	3.8158	1.7857	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	57.0507	54.2484	60.1584	58.3289	57.7500	38.8158	57.1429	86.0000	85.4211	-	-	-	-	-	-	-	-	-
RTE	58.2697	52.9412	64.7910	57.5132	55.1184	61.5789	85.7143	87.0000	83.5658	90.6137	-	-	-	-	-	-	-	-
IMDB	58.2656	51.7716	66.6224	51.3289	51.9474	61.8026	83.9286	89.0000	84.1316	89.8917	82.6316	-	-	-	-	-	-	-
SST-2	58.8497	53.4572	65.4522	48.6974	46.3421	62.7632	85.7143	88.0000	82.5263	88.4477	82.2632	94.9541	-	-	-	-	-	-
dbpedia	57.8783	52.7692	64.0827	46.7368	43.2763	63.0000	85.7143	94.0000	82.5658	86.2816	48.8158	60.4358	99.1579	-	-	-	-	-
agnews	59.8917	51.1868	72.1640	48.6053	50.7237	62.2895	83.9286	85.0000	77.7237	85.5596	81.2105	94.3807	88.2632	93.0395	-	-	-	-
yahoo	52.1987	47.6436	57.7168	43.4342	26.0789	62.2632	85.7143	35.0000	39.6842	66.4260	56.0658	77.1789	72.6316	85.1316	73.9868	-	-	-
MultiRC	60.1670	53.9044	68.0761	47.3553	45.5789	64.1711	87.5000	79.0000	83.3947	85.1986	80.0132	93.1193	91.7895	74.8289	48.9211	75.5569	-	-
BoolQA	59.8226	55.2804	65.1781	48.6842	41.2500	57.7763	64.2857	73.0000	79.6974	83.8989	79.2105	92.6606	89.6579	71.2105	37.5263	72.5248	84.9847	-
WiC	46.0985	51.0492	42.0231	24.6842	13.3158	28.4737	42.8571	37.0000	63.4737	19.4946	56.5658	42.0872	59.8947	45.7500	15.4079	65.4084	69.6024	74.2947

TABLE 10: The task-wise performance (%) of Baseline^R+KL under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.3784	63.1235	65.6842	65.6842	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.1231	62.0227	64.2632	64.5921	63.9342	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	62.1276	59.8555	64.5789	55.9868	52.7237	85.0263	-	-	-	-	-	-	-	-	-	-	-	-
CB	58.7442	59.9587	57.5779	50.6974	45.8816	75.8816	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	42.4836	59.5803	33.0110	50.8816	47.4342	0.1053	0.0000	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	58.3575	53.6636	63.9514	60.4342	53.7500	55.7105	80.3571	76.0000	85.6316	-	-	-	-	-	-	-	-	-
RTE	58.1002	52.1156	65.6375	60.3421	53.6184	62.9605	87.5000	85.0000	84.3158	90.2527	-	-	-	-	-	-	-	-
IMDB	60.5976	54.1796	68.7404	55.4868	56.4474	63.5000	87.5000	83.0000	84.3947	89.5307	82.7895	-	-	-	-	-	-	-
SST-2	60.2305	54.6956	67.0118	50.7500	50.2895	63.5263	87.5000	89.0000	84.1184	89.1697	81.8947	95.1835	-	-	-	-	-	-
dbpedia	60.1338	53.4228	68.7731	49.9079	46.4474	64.4737	87.5000	94.0000	84.6184	88.0866	65.3158	82.3394	99.1447	-	-	-	-	-
agnews	60.8271	52.0124	73.2392	52.6842	46.3816	64.1974	87.5000	82.0000	83.5526	83.0325	81.9605	94.9541	87.6842	93.1447	-	-	-	-
yahoo	46.7187	45.5452	47.9543	22.3421	2.8684	50.6974	60.7143	25.0000	35.9868	56.3177	39.4737	62.2706	74.0000	83.3158	73.2105	-	-	-
MultiRC	55.4975	56.8627	54.1962	41.7368	33.8684	29.7895	30.3571	76.0000	84.3026	35.3791	75.5395	92.6606	72.5921	53.3026	25.1711	75.2475	-	-
BoolQA	51.6320	56.6219	47.4503	38.6842	22.7105	4.1579	1.7857	45.0000	81.0132	1.0830	62.7895	81.7661	68.6974	52.5658	15.0395	73.2673	83.7003	-
WiC	42.4638	52.7692	35.5259	17.8026	7.2368	1.1974	0.0000	34.0000	60.4737	0.3610	48.9868	50.0000	51.0395	45.0000	15.4474	66.1304	69.3884	69.2790

TABLE 11: The task-wise performance (%) of Baseline^R+KL with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.3544	64.9123	65.8026	65.8026	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.3547	62.9515	63.7632	63.8421	63.6842	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.8955	61.4035	71.0965	64.9605	62.8289	85.5000	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.4280	62.1603	71.3248	65.1053	63.2105	85.4868	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	55.2835	59.2363	51.8252	62.6447	61.8553	30.5395	33.9286	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	63.2466	57.3443	70.5033	63.4211	62.1579	69.8421	89.2857	91.0000	86.1842	-	-	-	-	-	-	-	-	-
RTE	62.3130	57.0691	68.6180	64.1974	61.3421	62.5789	87.5000	93.0000	85.1053	90.2527	-	-	-	-	-	-	-	-
IMDB	62.8281	57.3443	69.4715	56.7500	58.3684	63.1579	87.5000	94.0000	85.1447	89.8917	82.7368	-	-	-	-	-	-	-
SST-2	63.8494	58.4451	70.3549	58.1053	57.9342	64.0132	89.2857	95.0000	85.2500	88.4477	82.3553	96.4450	-	-	-	-	-	-
dbpedia	65.5645	58.5827	74.4356	55.1316	56.2368	64.5789	87.5000	95.0000	85.6053	86.6426	82.6579	95.6422	99.1579	-	-	-	-	-
agnews	65.8130	57.8259	76.3600	56.2632	56.2632	64.6842	89.2857	97.0000	85.5000	86.2816	82.6053	95.5275	93.3026	92.9737	-	-	-	-
yahoo	60.6606	53.6292	69.8140	49.8421	42.7368	64.5658	91.0714	96.0000	70.5263	76.1733	80.1184	93.9220	87.4211	87.2763	72.5263	-	-	-
MultiRC	64.4890	57.4475	73.4978	55.6053	56.4605	64.7237	91.0714	97.0000	84.5658	84.8375	82.3816	95.2982	98.6184	86.5526	54.6053	75.2475	-	-
BoolQA	64.8161	58.4795	72.6927	55.7368	56.5263	65.0263	89.2857	98.0000	81.5789	81.5884	82.4211	95.1835	98.6579	85.8026	46.3684	74.4431	84.1896	-
WiC	60.3147	54.9364	66.8605	56.8289	53.1711	59.8947	82.1429	89.0000	67.5000	71.8412	81.3158	89.2202	97.1053	79.0658	33.3026	67.3267	73.1193	73.5110

TABLE 12: The task-wise performance (%) of Baseline^R+L1 under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.1065	64.2243	66.0132	66.0132	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.3069	62.7107	63.9145	64.2237	63.6053	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	63.4374	57.7227	70.4079	64.7237	61.2237	85.2763	-	-	-	-	-	-	-	-	-	-	-	-
CB	62.6949	56.6563	70.1741	64.6842	62.1711	83.5263	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	44.5302	56.2779	36.8400	56.9605	53.0263	0.0000	0.0000	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	58.2426	52.1844	65.8921	60.2237	59.4737	57.6053	83.9286	92.0000	85.7895	-	-	-	-	-	-	-	-	-
RTE	56.6823	49.0196	67.1845	61.1711	59.5395	62.6974	80.3571	88.0000	84.0658	91.6968	-	-	-	-	-	-	-	-
IMDB	55.7876	48.1596	66.2868	49.2763	50.5526	63.2632	83.9286	89.0000	84.6579	92.0578	82.3158	-	-	-	-	-	-	-
SST-2	57.0907	50.0860	66.3732	48.2237	49.3553	63.8289	85.7143	91.0000	84.3289	90.9747	81.5132	94.7248	-	-	-	-	-	-
dbpedia	62.7512	56.1404	71.1267	46.3947	48.1974	65.0132	89.2857	96.0000	84.5921	87.7256	79.6842	94.0367	99.1842	-	-	-	-	-
agnews	63.3294	55.4180	73.8758	49.3947	52.5526	64.6579	85.7143	97.0000	84.5132	81.9495	82.3684	94.7248	87.5263	93.0395	-	-	-	-
yahoo	56.9157	50.4988	65.2009	45.4474	32.3421	62.5000	85.7143	93.0000	59.4737	67.8700	74.9737	92.3165	83.5789	86.2632	73.3026	-	-	-
MultiRC	63.4436	57.0003	71.5293	52.2500	54.4737	64.6184	89.2857	97.0000	83.8947	83.7545	81.8816	95.4128	97.4474	80.9342	50.1711	76.0932	-	-
BoolQA	64.0448	57.7915	71.8155	52.9605	54.3684	64.4737	85.7143	98.0000	82.5921	82.3105	82.0263	95.1835	98.1316	81.6842	47.4737	75.3094	83.6086	-
WiC	60.0691	54.5924	66.7673	57.5395	54.6316	51.1316	80.3571	83.0000	67.8421	68.2310	80.6316	90.0229	95.1842	79.4868	39.7500	69.3276	73.1804	72.8840

TABLE 13: The task-wise performance (%) of Baseline^R+L1 with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.9877	64.6371	65.3421	65.3421	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.7353	63.8459	63.6250	63.9342	63.3158	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.0210	59.6491	71.4561	65.0263	63.1579	86.1842	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.5054	60.8187	70.9748	65.1184	62.9474	84.6842	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	55.3894	59.1675	52.0648	62.9605	61.6316	31.1579	30.3571	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	63.0276	57.6883	69.4561	63.4868	61.2368	66.7237	85.7143	93.0000	85.9474	-	-	-	-	-	-	-	-	-
RTE	61.9077	57.2067	67.4505	63.8816	59.6184	61.7237	87.5000	95.0000	83.1316	93.1408	-	-	-	-	-	-	-	-
IMDB	62.5765	57.8947	68.0821	55.1053	55.3289	62.5395	87.5000	95.0000	83.5000	91.6968	82.5789	-	-	-	-	-	-	-
SST-2	62.9953	58.1699	68.6936	54.7237	54.9079	63.5263	89.2857	96.0000	83.8289	90.6137	82.0526	95.8716	-	-	-	-	-	-
dbpedia	64.5297	58.2043	72.3974	52.0789	50.1579	64.3947	91.0714	94.0000	84.2763	86.6426	80.8421	94.7248	99.1316	-	-	-	-	-
agnews	65.1227	57.5851	74.9307	53.9474	54.2237	64.5658	87.5000	95.0000	83.9211	81.9495	82.6053	95.5275	89.5658	92.7105	-	-	-	-
yahoo	58.8319	53.6292	65.1526	47.0526	33.7237	60.7105	83.9286	87.0000	55.3421	62.8159	78.0526	92.4312	81.8684	87.5789	73.4211	-	-	-
MultiRC	63.7090	56.5875	72.8810	52.2500	53.1842	64.4211	89.2857	97.0000	84.3816	84.8375	82.2632	95.2982	97.7105	88.0526	55.4211	75.8870	-	-
BoolQA	64.2837	58.3075	71.6247	54.8684	55.4079	64.3289	85.7143	95.0000	81.4342	82.3105	82.4342	95.4128	97.8816	84.7105	40.6842	74.6081	85.1376	-
WiC	60.6113	55.0052	67.4899	56.1842	54.8026	59.0921	75.0000	91.0000	65.4737	71.1191	82.2105	94.1514	95.9737	82.5000	34.5789	68.4406	77.6147	74.4514

TABLE 14: The task-wise performance (%) of Baseline^R+L2 under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.9750	64.1555	65.8158	65.8158	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.8210	63.7083	63.9342	63.9342	63.8421	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.9386	62.2291	70.1184	64.8684	60.8684	84.6184	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.0262	60.8187	69.8591	65.7105	61.1579	82.5132	96.4286	-	-	-	-	-	-	-	-	-	-	-
COPA	50.7358	59.8555	44.0277	61.6711	60.5395	9.4737	3.5714	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	62.3038	56.6219	69.2532	63.4868	61.2895	65.8289	83.9286	91.0000	86.0132	-	-	-	-	-	-	-	-	-
RTE	61.5376	56.1404	68.0829	63.6974	60.0921	62.8947	87.5000	93.0000	84.3816	89.8917	-	-	-	-	-	-	-	-
IMDB	61.4934	57.1723	66.5210	49.6447	51.6053	63.1184	89.2857	93.0000	84.3026	90.9747	82.5263	-	-	-	-	-	-	-
SST-2	63.0449	59.2707	67.3324	50.2500	52.0921	63.5658	91.0714	95.0000	84.3684	88.8087	81.9474	94.4954	-	-	-	-	-	-
dbpedia	64.8693	58.3419	73.0413	51.4605	53.2500	64.5921	91.0714	96.0000	84.7237	83.0325	81.7105	94.7248	99.2237	-	-	-	-	-
agnews	64.5281	57.3443	73.7694	53.0658	53.7895	63.4605	87.5000	94.0000	81.7368	79.4224	82.5526	95.2982	85.7368	93.0000	-	-	-	-
yahoo	55.2611	52.0468	58.8986	47.2895	27.2895	61.3947	78.5714	89.0000	34.8684	66.7870	61.5526	85.4358	78.0000	83.7895	73.1316	-	-	-
MultiRC	64.8105	59.8211	70.7078	54.4211	54.2763	64.2105	87.5000	95.0000	83.2632	82.3105	82.0395	94.2661	96.4079	79.8816	44.7105	75.2269	-	-
BoolQA	64.1808	59.6147	69.5043	56.1711	55.9737	60.9474	75.0000	97.0000	82.8816	75.4513	82.2105	94.7248	95.9079	77.0000	33.6842	73.2467	81.9572	-
WiC	61.2686	56.6219	66.7462	56.7237	54.3947	56.6711	85.7143	92.0000	65.4474	66.4260	81.5395	93.3486	93.8026	78.7895	37.1974	67.6155	77.6147	74.6082

TABLE 15: The task-wise performance (%) of Baseline^R+L2 with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.6759	66.0131	65.3421	65.3421	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.7975	65.0843	64.5132	64.8421	64.1842	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.9683	64.7059	69.3947	62.9474	60.8684	84.3684	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.9523	64.8091	69.2422	62.8553	60.9474	83.7763	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	65.2217	63.6395	66.8845	63.0132	61.5000	75.6184	87.5000	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	67.0591	64.4307	69.9110	62.7500	60.3289	71.1711	87.5000	97.0000	84.9079	-	-	-	-	-	-	-	-	-
RTE	64.1356	62.2979	66.0850	61.8684	59.4605	61.4605	85.7143	96.0000	80.1842	88.8087	-	-	-	-	-	-	-	-
IMDB	64.5184	61.6443	67.6736	56.6711	53.8947	63.0789	89.2857	94.0000	81.3026	85.1986	82.2763	-	-	-	-	-	-	-
SST-2	65.5431	61.9195	69.6171	58.2500	57.1579	63.7632	89.2857	94.0000	82.6579	84.4765	82.2763	95.5275	-	-	-	-	-	-
dbpedia	66.6747	60.5091	74.2394	56.6974	55.2368	66.9605	91.0714	90.0000	82.5000	80.8664	82.0789	95.0688	99.0000	-	-	-	-	-
agnews	68.2080	61.7475	76.1783	57.1184	53.6447	64.9605	91.0714	85.0000	83.2237	80.8664	82.7105	95.5275	96.6447	92.3289	-	-	-	-
yahoo	66.5029	59.9243	74.7041	56.3421	51.7763	64.7895	91.0714	89.0000	83.3421	78.7004	81.9211	94.2661	95.3684	90.1184	71.2763	-	-	-
MultiRC	67.5049	60.9907	75.5769	56.7368	55.9474	64.7500	87.5000	93.0000	83.1842	80.8664	82.4868	95.0688	96.9868	89.6053	71.1974	77.1040	-	-
BoolQA	63.1795	56.1404	72.2370	55.5658	56.0658	51.6316	53.5714	69.0000	68.5789	59.5668	81.5658	95.4128	96.8553	89.3684	71.2895	71.0190	85.5657	-
WiC	64.3973	57.9635	72.4376	56.5263	55.5000	61.2105	76.7857	81.0000	62.9342	68.9531	81.6447	95.1835	98.0263	88.0000	70.0921	72.1328	78.9602	76.3323

TABLE 16: The task-wise performance (%) of Baseline^R+L1 (w=100) under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	66.3682	66.4603	66.2763	66.2763	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.9774	65.1531	64.8026	65.0000	64.6053	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.9586	64.7403	67.2237	59.8026	58.0263	83.8421	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.5301	64.3963	66.7046	60.2500	58.0395	81.6316	92.8571	-	-	-	-	-	-	-	-	-	-	-
COPA	62.8714	62.8139	62.9291	59.5526	57.2895	71.3158	87.5000	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	64.7986	63.4675	66.1867	55.1974	52.2368	71.9342	92.8571	95.0000	84.8026	-	-	-	-	-	-	-	-	-
RTE	63.7388	62.8483	64.6548	56.9474	55.0132	61.5789	91.0714	95.0000	83.6974	86.2816	-	-	-	-	-	-	-	-
IMDB	64.6181	62.9515	66.3753	51.2237	50.9342	62.2237	91.0714	93.0000	83.7895	85.9206	82.4605	-	-	-	-	-	-	-
SST-2	65.4947	62.3667	68.9531	56.0395	55.2763	63.0132	91.0714	93.0000	84.2632	87.3646	81.9605	95.6422	-	-	-	-	-	-
dbpedia	66.9472	60.6467	74.7085	58.0526	56.8158	64.0789	91.0714	92.0000	84.4868	86.6426	82.5658	95.8716	99.0395	-	-	-	-	-
agnews	68.2635	61.4723	76.7416	57.6447	54.8026	64.5000	91.0714	96.0000	84.2632	86.6426	82.6447	96.5596	98.3684	91.9737	-	-	-	-
yahoo	67.7980	61.8851	74.9601	54.4474	52.2500	65.3816	89.2857	95.0000	83.6711	86.2816	82.4079	96.4450	97.5921	89.4474	71.2368	-	-	-
MultiRC	68.0148	62.2979	74.8869	53.8289	51.4737	65.9474	85.7143	96.0000	83.7368	83.3935	81.9079	95.6422	97.6316	89.4211	71.3026	76.1345	-	-
BoolQA	68.6232	62.5387	76.0193	58.9737	54.8947	68.5921	82.1429	96.0000	82.0658	81.2274	82.1316	94.9541	97.9868	87.8421	70.9868	73.8449	83.9144	-
WiC	64.5103	59.0299	71.1125	55.4211	51.0132	63.9079	78.5714	95.0000	62.7368	72.5632	78.6579	88.6468	98.3421	88.7105	71.1711	65.2021	71.2844	73.5110

TABLE 17: The task-wise performance (%) of Baseline^R+L1 (w=100) with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.7533	65.8755	65.6316	65.6316	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.7048	65.1187	64.2961	64.1053	64.4868	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.8373	64.1211	69.7939	63.5132	60.4474	85.4211	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.9012	64.2243	69.8110	63.3158	60.6579	85.3158	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	60.6423	63.2955	58.2026	62.5395	60.6447	51.1316	28.5714	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.4575	61.6099	72.1331	62.4211	59.5263	80.2105	89.2857	94.0000	85.9605	-	-	-	-	-	-	-	-	-
RTE	62.4516	59.9587	65.1607	60.6447	56.4605	62.1974	80.3571	90.0000	80.0000	89.8917	-	-	-	-	-	-	-	-
IMDB	62.4123	60.1307	64.8739	45.9211	47.3684	64.3684	89.2857	96.0000	82.6579	86.2816	82.6842	-	-	-	-	-	-	-
SST-2	64.4296	60.3715	69.0726	55.6316	55.3026	64.8026	91.0714	95.0000	83.5263	86.2816	81.9605	95.2982	-	-	-	-	-	-
dbpedia	65.7177	58.9267	74.2778	57.0526	54.8158	65.3684	91.0714	93.0000	83.4079	84.8375	82.6184	96.1009	99.1447	-	-	-	-	-
agnews	67.2053	60.5779	75.4610	53.3421	50.1842	65.3158	91.0714	92.0000	83.2500	84.4765	82.5658	95.8716	97.9737	92.5921	-	-	-	-
yahoo	65.6775	59.4771	73.3210	51.6974	46.6579	65.4868	89.2857	91.0000	83.0000	82.6715	82.5395	95.1835	92.1447	89.8158	72.0263	-	-	-
MultiRC	67.7128	61.7819	74.9033	54.7105	50.5395	65.8026	91.0714	97.0000	83.8947	85.5596	82.6316	95.7569	96.9211	88.2237	71.7105	77.4134	-	-
BoolQA	67.8075	61.6787	75.2887	55.8816	52.4342	65.5395	85.7143	96.0000	82.2895	77.6173	82.2763	95.7569	98.3553	88.3026	69.8553	74.9587	86.4526	-
WiC	66.1531	60.7499	72.6112	54.8553	50.4474	62.2105	75.0000	95.0000	69.0395	70.7581	82.2237	94.6101	98.1053	88.7237	70.8026	70.1526	79.6636	75.5486

TABLE 18: The task-wise performance (%) of Baseline^R+L2 (w=100) under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.7186	65.9787	65.4605	65.4605	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.9316	65.6691	64.2105	64.6053	63.8158	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	63.5951	64.6027	62.6184	53.8684	48.1842	85.8026	-	-	-	-	-	-	-	-	-	-	-	-
CB	63.7175	64.6371	62.8238	56.4079	50.9079	80.9211	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	32.8576	61.4379	22.4255	36.8947	29.5658	0.0000	0.0000	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	63.9195	61.2315	66.8543	64.3421	61.0526	56.2500	51.7857	90.0000	85.5789	-	-	-	-	-	-	-	-	-
RTE	61.8861	57.6539	66.7888	63.3684	57.4605	62.4868	83.9286	95.0000	82.5658	88.0866	-	-	-	-	-	-	-	-
IMDB	62.1552	57.6883	67.3718	56.1447	54.9737	60.3553	83.9286	93.0000	81.9342	88.4477	82.2237	-	-	-	-	-	-	-
SST-2	62.9882	57.5851	69.5102	58.0789	57.1316	63.2895	87.5000	96.0000	83.2105	87.7256	81.7237	95.4128	-	-	-	-	-	-
dbpedia	65.7139	58.2731	75.3331	59.0658	57.7237	66.1579	87.5000	95.0000	84.6842	86.6426	82.2237	95.6422	99.0526	-	-	-	-	-
agnews	65.6319	57.1723	77.0296	58.1711	55.7500	65.8947	89.2857	97.0000	84.8553	85.1986	82.5132	95.6422	96.5263	92.7105	-	-	-	-
yahoo	66.0016	59.4427	74.1873	53.3158	49.1184	64.8947	87.5000	96.0000	84.3947	85.9206	82.2500	94.8394	93.4868	89.3158	71.5395	-	-	-
MultiRC	66.4442	60.0275	74.3970	53.8553	50.9868	67.7632	85.7143	98.0000	84.2105	83.7545	79.9342	95.1835	93.0658	89.8684	71.1184	76.3614	-	-
BoolQA	66.9701	59.7179	76.2272	58.1053	56.3684	68.7500	85.7143	96.0000	81.5789	79.4224	81.7105	95.7569	96.7632	89.2237	71.4211	74.5875	86.1162	-
WiC	64.6643	57.8947	73.2265	56.9737	54.8158	67.3026	85.7143	97.0000	64.1316	73.2852	80.5263	93.2339	96.5395	89.7763	71.3289	71.1427	79.6942	76.3323

TABLE 19: The task-wise performance (%) of Baseline^R+L2 (w=100) with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.8252	65.7035	65.9474	65.9474	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.7036	65.0155	64.3947	64.9868	63.8026	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.9695	62.7795	71.7588	65.2105	64.1579	85.9079	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.3947	62.0227	71.4298	65.1842	64.4211	84.5132	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	51.3578	59.8555	44.9730	63.7500	63.3158	7.4868	0.0000	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.0079	60.8531	72.1168	64.2895	63.2763	74.3158	85.7143	98.0000	86.1447	-	-	-	-	-	-	-	-	-
RTE	64.3613	60.4059	68.8710	63.8816	62.2895	62.9605	87.5000	98.0000	85.0000	91.6968	-	-	-	-	-	-	-	-
IMDB	64.2449	59.8211	69.3753	57.0132	57.4737	63.6316	91.0714	98.0000	84.5789	90.6137	82.8684	-	-	-	-	-	-	-
SST-2	65.1182	60.1995	70.9121	59.1974	59.5921	64.3421	89.2857	98.0000	84.8026	87.7256	82.6579	95.8716	-	-	-	-	-	-
dbpedia	61.1287	53.1476	71.9305	53.7500	51.1711	65.0526	87.5000	95.0000	85.3684	80.8664	74.0658	91.7431	99.1579	-	-	-	-	-
agnews	64.1442	55.2116	76.5251	59.4211	58.7632	65.5921	89.2857	98.0000	83.4342	72.2022	82.7763	94.4954	90.4211	92.9868	-	-	-	-
yahoo	54.1623	47.5404	62.9273	55.8553	37.2632	59.0789	69.6429	59.0000	27.5000	45.8484	79.1711	91.1697	80.9079	87.6579	73.3684	-	-	-
MultiRC	62.6109	55.5556	71.7190	56.5395	56.9868	66.2368	87.5000	84.0000	84.0658	81.5884	82.5000	94.7248	96.8553	75.8947	48.9868	75.4950	-	-
BoolQA	63.5635	56.5187	72.6144	57.4868	58.1447	66.5395	87.5000	63.0000	84.2368	82.6715	82.7763	95.6422	97.9737	77.2368	46.1974	75.3507	85.5963	-
WiC	59.4309	53.1132	67.4546	57.3158	55.1053	60.9079	80.3571	86.0000	60.9605	68.9531	82.1316	91.5138	97.7237	77.8158	47.6579	60.3960	68.8991	76.3323

TABLE 20: The task-wise performance (%) of Baseline^R+L1 (w=dy) under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.7648	65.3251	66.2105	66.2105	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.7685	63.7083	63.8289	64.4079	63.2500	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	64.3991	61.2315	67.9123	61.3421	57.0789	85.3158	-	-	-	-	-	-	-	-	-	-	-	-
CB	63.8369	60.6811	67.3390	61.7763	57.7500	82.2763	96.4286	-	-	-	-	-	-	-	-	-	-	-
COPA	9.8956	58.8923	5.4016	11.6974	3.3421	0.0000	0.0000	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	62.0373	56.2092	69.2139	64.6579	61.4868	64.6447	87.5000	92.0000	85.6316	-	-	-	-	-	-	-	-	-
RTE	58.2216	53.8700	63.3380	63.5789	58.4737	62.1974	87.5000	92.0000	67.5658	90.2527	-	-	-	-	-	-	-	-
IMDB	60.9780	55.5900	67.5227	53.0658	55.2500	62.7500	87.5000	95.0000	82.3947	88.8087	82.8684	-	-	-	-	-	-	-
SST-2	61.6219	56.6907	67.4927	51.5921	52.6184	63.8421	91.0714	97.0000	83.0132	87.0036	81.9868	94.8394	-	-	-	-	-	-
dbpedia	59.2123	52.1844	68.4277	43.3421	44.7105	64.6447	89.2857	94.0000	83.1053	83.0325	72.3421	87.7294	99.1842	-	-	-	-	-
agnews	60.6088	52.3564	71.9494	43.4211	47.8816	63.8158	85.7143	93.0000	82.2368	83.3935	82.3289	95.1835	87.4868	93.0132	-	-	-	-
yahoo	47.1703	40.9013	55.7089	35.8816	20.7368	61.5789	76.7857	80.0000	9.5395	62.4549	70.3684	89.7936	81.3289	88.1579	73.4474	-	-	-
MultiRC	59.5038	53.4228	67.1471	47.6447	49.9605	64.7237	91.0714	20.0000	84.7500	86.2816	81.2368	94.6101	95.2763	70.4737	34.7368	74.9381	-	-
BoolQA	61.0059	55.4524	67.7955	48.4474	49.8947	64.6711	91.0714	14.0000	83.7895	86.6426	82.0658	95.1835	95.9605	70.1974	33.0132	73.2467	85.3517	-
WiC	58.3556	53.2852	64.4925	49.4079	50.3026	62.8684	85.7143	22.0000	71.2368	81.2274	81.6316	94.6101	93.7237	68.3421	30.3816	65.2847	71.7431	73.5110

TABLE 21: The task-wise performance (%) of Baseline^R+L1 (w=dy) with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	66.3052	66.1507	66.4605	66.4605	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.8726	64.6027	65.1447	65.4737	64.8158	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.7933	63.1579	70.8728	64.1711	63.6316	84.8158	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.7035	62.9515	70.9310	64.4342	63.4605	84.7237	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	49.6547	59.8555	42.4246	63.1974	62.5526	1.0921	3.5714	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	63.6291	57.3099	71.5146	62.9737	61.9868	74.7763	89.2857	96.0000	85.8684	-	-	-	-	-	-	-	-	-
RTE	61.9088	57.0347	67.6937	63.0000	60.3684	62.1711	83.9286	97.0000	83.9868	88.0866	-	-	-	-	-	-	-	-
IMDB	61.9748	57.1723	67.6580	53.2763	53.6974	63.3289	87.5000	97.0000	84.1184	87.0036	82.6316	-	-	-	-	-	-	-
SST-2	63.2228	57.9979	69.4823	56.1447	55.8553	64.0132	85.7143	96.0000	84.6184	87.0036	82.6316	95.9862	-	-	-	-	-	-
dbpedia	65.5600	58.7203	74.2032	54.5658	56.3289	64.7368	89.2857	96.0000	84.9737	82.3105	82.2632	95.4128	99.2237	-	-	-	-	-
agnews	64.6367	56.2092	76.0371	57.6184	55.6316	64.7237	87.5000	95.0000	84.7368	79.0614	82.6974	95.4128	91.2632	92.9211	-	-	-	-
yahoo	56.9917	51.4964	63.8000	51.8947	35.6974	60.9737	80.3571	78.0000	46.4342	49.4585	72.8947	91.8578	80.5658	86.3816	72.5526	-	-	-
MultiRC	66.5245	59.8555	74.8660	58.2368	57.4605	64.7237	85.7143	97.0000	84.5132	81.5884	82.3026	95.4128	98.2105	83.7895	65.8684	76.1964	-	-
BoolQA	65.9936	58.7891	75.2104	59.4342	58.2368	64.6974	87.5000	96.0000	81.8421	80.1444	82.4079	95.4128	98.2368	84.7895	64.5789	75.5363	85.4128	-
WiC	62.1001	55.1772	71.0094	58.9605	55.4211	58.7368	71.4286	94.0000	63.8026	63.5379	81.9342	93.2339	98.3026	84.3289	61.7105	70.8127	75.9939	74.2947

TABLE 22: The task-wise performance (%) of Baseline^R+L2 (w=dy) under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.7911	65.4283	66.1579	66.1579	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.3645	62.9515	63.7829	64.0658	63.5000	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	67.1038	63.7083	70.8816	65.2763	62.0000	85.3684	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.9670	62.9515	69.2860	64.7237	61.6579	81.2632	98.2143	-	-	-	-	-	-	-	-	-	-	-
COPA	46.6359	58.6859	38.6914	60.0789	55.5132	0.0000	0.0000	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	61.3321	55.8996	67.9343	64.5395	63.0263	57.8289	62.5000	92.0000	86.0658	-	-	-	-	-	-	-	-	-
RTE	59.3602	52.3564	68.5272	64.0000	61.6316	62.6447	83.9286	89.0000	84.7500	87.7256	-	-	-	-	-	-	-	-
IMDB	58.6381	51.8060	67.5461	52.8816	54.0658	62.3947	91.0714	88.0000	84.6184	87.7256	82.5921	-	-	-	-	-	-	-
SST-2	60.2293	53.3540	69.1388	55.1184	56.0263	63.8289	87.5000	90.0000	84.4868	86.6426	82.1711	95.4128	-	-	-	-	-	-
dbpedia	62.2228	53.4572	74.4270	55.0395	57.2105	65.0395	87.5000	95.0000	85.0789	83.7545	82.0132	95.1835	99.0921	-	-	-	-	-
agnews	63.7946	54.9708	75.9930	55.6053	58.0658	64.7763	85.7143	95.0000	85.1447	82.6715	82.5921	95.6422	89.4605	93.4868	-	-	-	-
yahoo	58.2773	50.4300	69.0170	54.0789	44.6184	64.5658	87.5000	94.0000	66.1053	75.4513	72.1579	93.1193	84.4211	89.2763	73.4474	-	-	-
MultiRC	64.7834	57.4475	74.2670	57.2632	56.7500	64.6316	85.7143	84.0000	82.6053	75.4513	81.8947	94.8394	98.3289	89.0658	61.1711	73.9686	-	-
BoolQA	63.4517	56.3123	72.6642	56.1316	56.6053	63.3026	82.1429	88.0000	81.3421	71.8412	82.3026	95.5275	98.0263	89.1316	46.7105	72.4010	84.4343	-
WiC	60.7927	54.7988	68.2590	55.7763	53.4868	57.9079	82.1429	65.0000	67.5921	64.9819	81.2895	93.4633	96.0526	89.3553	39.9868	65.2021	75.6881	74.7649

TABLE 23: The task-wise performance (%) of Baseline^R+L2 (w=dy) with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.4521	65.0155	65.8947	65.8947	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.8612	63.8803	63.8421	63.9868	63.6974	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.4479	61.3003	70.1974	62.5658	61.6316	86.3947	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.3685	60.9219	70.5154	63.5000	61.9079	86.0000	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	52.6889	58.0667	48.2227	60.8947	60.3816	23.0263	10.7143	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	63.0186	59.6147	66.8347	61.3553	59.4605	59.8421	80.3571	92.0000	86.2500	-	-	-	-	-	-	-	-	-
RTE	62.5993	58.5483	67.2526	62.0526	58.6842	62.3553	82.1429	96.0000	84.5658	90.9747	-	-	-	-	-	-	-	-
IMDB	63.6348	59.1331	68.8783	56.4737	56.3816	62.9342	85.7143	97.0000	84.6974	89.8917	82.6447	-	-	-	-	-	-	-
SST-2	64.0274	59.1675	69.7570	56.9868	56.8289	63.6711	85.7143	95.0000	84.8421	88.8087	82.4079	95.0688	-	-	-	-	-	-
dbpedia	66.3010	60.0963	73.9345	54.2895	54.7237	64.3421	85.7143	97.0000	85.4342	87.0036	82.4474	94.7248	99.1184	-	-	-	-	-
agnews	66.6486	59.1331	76.3526	57.1053	56.8947	64.6053	85.7143	95.0000	85.3816	85.9206	82.6711	95.2982	91.6974	93.2763	-	-	-	-
yahoo	63.0513	55.6244	72.7671	51.4474	51.6974	64.6579	85.7143	93.0000	81.2632	79.7834	81.7895	94.8394	87.3553	87.2895	73.4868	-	-	-
MultiRC	65.2985	57.5851	75.3977	55.1974	55.5395	64.4474	82.1429	96.0000	84.9079	87.0036	82.4342	95.0688	98.3684	87.1842	71.8816	75.7426	-	-
BoolQA	66.3967	59.1331	75.6946	55.8684	56.2763	65.0263	83.9286	97.0000	83.4737	80.8664	82.3421	95.0688	98.6447	87.0395	70.5395	75.0825	84.9541	-
WiC	61.9756	55.3836	70.3490	57.4868	54.5526	57.7763	76.7857	89.0000	63.1842	68.2310	80.6447	87.8440	98.5789	87.6184	69.1053	61.2624	63.2722	75.8621

TABLE 24: The task-wise performance (%) of Baseline^R+TM under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.1574	64.4995	65.8289	65.8289	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.1013	61.8163	64.4408	64.9605	63.9211	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	62.4140	57.4819	68.2719	61.7895	57.4737	85.5526	-	-	-	-	-	-	-	-	-	-	-	-
CB	63.0028	57.9635	69.0016	63.3289	59.3158	84.2237	87.5000	-	-	-	-	-	-	-	-	-	-	-
COPA	45.2508	58.9611	36.7137	56.6974	52.4605	0.4474	0.0000	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	61.5596	57.6539	66.0329	62.0526	60.5132	55.2895	83.9286	94.0000	85.7763	-	-	-	-	-	-	-	-	-
RTE	61.1728	55.7276	67.7975	63.3289	60.0395	62.6447	78.5714	95.0000	83.8289	92.7798	-	-	-	-	-	-	-	-
IMDB	61.3152	56.5531	66.9529	50.8947	52.3553	63.1974	78.5714	96.0000	84.4737	90.9747	82.5000	-	-	-	-	-	-	-
SST-2	62.2389	57.2755	68.1440	52.1842	53.5263	64.1184	80.3571	96.0000	84.5658	89.8917	81.8553	96.2156	-	-	-	-	-	-
dbpedia	64.5741	57.4475	73.7192	53.4737	55.8947	65.1842	85.7143	98.0000	84.7895	88.4477	80.3553	95.6422	99.1579	-	-	-	-	-
agnews	64.5775	56.5531	75.2555	53.9079	56.3947	64.3816	78.5714	99.0000	84.6184	88.4477	82.6316	95.1835	88.6711	93.0789	-	-	-	-
yahoo	58.1764	54.1796	62.8098	47.9079	28.8553	60.3947	80.3571	97.0000	51.6974	65.3430	71.5000	92.4312	83.0789	82.0263	72.9474	-	-	-
MultiRC	65.0772	59.1675	72.2985	56.0132	56.2763	61.2500	80.3571	98.0000	83.6579	79.7834	81.7237	95.0688	97.2105	82.4868	54.8684	74.8350	-	-
BoolQA	65.5071	59.9243	72.2370	56.4474	57.2632	58.9737	71.4286	99.0000	81.8289	75.4513	82.1316	95.7569	97.9342	83.3026	51.3947	73.0817	83.6697	-
WiC	62.7138	57.6539	68.7473	58.0263	54.3947	53.8158	83.9286	98.0000	66.9605	70.0361	81.6711	93.3486	97.2632	81.4868	49.9737	67.8012	75.9021	75.2351

TABLE 25: The task-wise performance (%) of Baseline^R+TM with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.1828	66.1851	64.2105	64.2105	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.2106	65.2219	63.2303	63.6316	62.8289	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	67.0835	64.5683	69.8026	63.6316	61.7368	84.0395	-	-	-	-	-	-	-	-	-	-	-	-
CB	67.1489	64.4995	70.0254	63.8553	62.0395	84.0395	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	67.3024	65.1531	69.5984	63.4079	61.6447	83.2632	89.2857	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	67.5704	64.9467	70.4150	61.0658	59.5658	76.6974	87.5000	95.0000	83.8816	-	-	-	-	-	-	-	-	-
RTE	65.2070	64.9123	65.5045	60.9605	57.7368	61.0921	83.9286	94.0000	80.9868	85.5596	-	-	-	-	-	-	-	-
IMDB	67.0415	64.8779	69.3545	59.8684	58.6184	63.6184	87.5000	94.0000	81.7895	88.4477	81.7237	-	-	-	-	-	-	-
SST-2	67.5522	64.9123	70.4160	60.3026	58.8553	64.7500	83.9286	95.0000	82.4342	86.2816	81.8553	95.5275	-	-	-	-	-	-
dbpedia	69.8301	64.1555	76.6059	63.4211	62.0658	66.9868	85.7143	93.0000	82.9737	84.1155	82.3684	96.1009	99.0263	-	-	-	-	-
agnews	70.3900	63.9491	78.2736	62.5658	60.9474	66.1579	87.5000	94.0000	83.1184	83.3935	82.6579	95.6422	98.9211	91.0921	-	-	-	-
yahoo	68.0177	62.6075	74.4513	57.7500	53.8158	64.5263	83.9286	93.0000	80.2368	81.5884	81.1974	94.6101	98.4079	86.8684	69.9211	-	-	-
MultiRC	67.9008	61.9883	75.0601	57.9211	56.0658	64.7368	85.7143	92.0000	80.9211	83.3935	81.0395	94.6101	98.5921	87.7237	69.9079	76.1964	-	-
BoolQA	68.2503	61.9883	75.9196	61.3684	57.8816	67.0789	83.9286	90.0000	77.9474	79.0614	81.7895	94.6101	98.7763	87.5789	70.0395	74.3399	83.8226	-
WiC	66.7359	60.7155	74.0817	59.8553	57.2368	60.1579	71.4286	88.0000	75.3289	76.5343	80.7105	94.1514	98.8421	86.6579	69.0263	74.9381	78.4404	72.4138

TABLE 26: The task-wise performance (%) of Baseline^R+TM (w=100) under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.6749	66.3915	64.9737	64.9737	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.9092	65.2563	64.5658	64.7895	64.3421	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.8225	65.2219	66.4342	57.8684	57.0132	84.4211	-	-	-	-	-	-	-	-	-	-	-	-
CB	64.7232	64.6715	64.7751	59.1711	57.8158	77.1184	94.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	50.8783	63.3643	42.5030	57.7895	54.6053	14.6579	7.1429	97.0000	-	-	-	-	-	-	-	-	-	-
QQP	65.1654	62.8827	67.6201	56.8158	54.8947	72.9868	91.0714	95.0000	85.2500	-	-	-	-	-	-	-	-	-
RTE	62.4651	62.4355	62.4947	56.0526	53.8158	61.1447	89.2857	89.0000	77.5263	87.0036	-	-	-	-	-	-	-	-
IMDB	64.9101	62.9171	67.0335	55.8289	56.0789	61.5263	89.2857	93.0000	78.1184	87.3646	82.3684	-	-	-	-	-	-	-
SST-2	65.6610	62.8827	68.6961	57.5132	57.4605	62.4342	89.2857	90.0000	79.9079	87.7256	81.8947	96.1009	-	-	-	-	-	-
dbpedia	68.5353	63.9491	73.8301	58.4342	56.7368	62.7500	89.2857	92.0000	80.3026	87.0036	82.4342	95.8716	98.9605	-	-	-	-	-
agnews	69.3027	63.6051	76.1215	57.2895	55.4211	63.2895	89.2857	90.0000	81.8026	87.0036	82.6711	96.3303	97.8947	91.4868	-	-	-	-
yahoo	68.2932	63.5363	73.8201	53.6447	51.8289	62.1053	91.0714	86.0000	81.8816	86.6426	82.4079	96.3303	95.9605	88.8026	70.5921	-	-	-
MultiRC	67.8287	62.6763	73.9041	50.9474	49.7632	65.2895	89.2857	95.0000	83.6053	84.8375	80.1579	93.8073	96.0263	89.2237	71.7632	76.0726	-	-
BoolQA	68.5678	62.6763	75.6818	55.8947	55.9868	65.0658	82.1429	95.0000	81.8684	75.0903	82.0132	95.2982	97.0395	89.0263	71.4605	75.9282	85.9327	-
WiC	66.2396	60.9907	72.4771	51.8158	52.5526	62.5132	82.1429	93.0000	72.4605	73.2852	80.3158	93.6927	98.1711	88.8553	71.6579	71.8647	70.1223	73.3542

TABLE 27: The task-wise performance (%) of Baseline^R+TM (w=100) with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	66.2360	66.0131	66.4605	66.4605	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	65.2747	66.0131	64.5526	64.5526	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	67.1812	63.7771	70.9693	64.9737	63.7632	84.1711	-	-	-	-	-	-	-	-	-	-	-	-
CB	67.4603	63.8459	71.5086	65.2105	64.0395	85.1316	91.0714	-	-	-	-	-	-	-	-	-	-	-
COPA	67.7171	64.1899	71.6545	65.3421	63.9342	85.2237	87.5000	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	68.4733	64.9811	72.3622	64.5526	62.3947	76.8289	87.5000	97.0000	85.2368	-	-	-	-	-	-	-	-	-
RTE	66.7533	64.9467	68.6634	64.5658	62.6053	62.5658	87.5000	97.0000	83.6579	89.1697	-	-	-	-	-	-	-	-
IMDB	67.1946	64.9123	69.6433	60.1842	57.8684	62.7895	89.2857	98.0000	83.8947	87.3646	82.3158	-	-	-	-	-	-	-
SST-2	67.8099	64.7747	71.1436	61.2763	59.7895	64.0526	89.2857	97.0000	84.1974	87.7256	82.5132	95.6422	-	-	-	-	-	-
dbpedia	68.1646	62.6763	74.7063	55.8026	55.4211	67.1974	89.2857	96.0000	84.8553	85.5596	82.6711	95.1835	99.1579	-	-	-	-	-
agnews	69.7987	63.1579	78.0002	59.8947	57.9342	66.2105	87.5000	96.0000	84.8816	83.3935	82.7763	94.8394	99.0000	92.8684	-	-	-	-
yahoo	66.5175	60.3371	74.1084	53.0263	49.8816	64.2105	87.5000	91.0000	83.0658	73.2852	82.5658	94.7248	96.9079	87.7632	72.7895	-	-	-
MultiRC	68.4534	62.2635	76.0100	59.6053	57.8684	65.5921	85.7143	92.0000	83.8289	82.6715	82.4211	95.0688	98.6184	87.0132	69.9342	76.7739	-	-
BoolQA	68.4335	61.8163	76.6373	61.4868	59.0921	66.3421	85.7143	94.0000	82.6316	79.4224	82.6184	95.9862	98.7895	86.8684	69.5263	75.5982	85.4434	-
WiC	66.9386	60.7843	74.4796	60.9474	59.1579	62.3684	82.1429	94.0000	71.2763	73.2852	82.2763	94.3807	98.7763	86.9474	69.6842	72.4010	81.8960	74.2947

TABLE 28: The task-wise performance (%) of Baseline^R+TM (w=dy) under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	66.0494	65.7723	66.3289	66.3289	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.0687	63.4331	64.7171	64.7632	64.6711	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	67.5616	64.6371	70.7632	65.3816	62.2500	84.6579	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.9329	64.0179	70.1260	64.9737	62.1842	83.0132	98.2143	-	-	-	-	-	-	-	-	-	-	-
COPA	49.1982	58.4795	42.4595	60.0000	59.6974	7.2237	8.9286	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	65.3409	61.7819	69.3350	64.3421	62.7500	63.8421	62.5000	97.0000	86.0921	-	-	-	-	-	-	-	-	-
RTE	63.4401	59.8211	67.5251	62.7763	59.5658	63.1053	89.2857	96.0000	83.3158	89.5307	-	-	-	-	-	-	-	-
IMDB	64.3878	60.3371	69.0214	56.2895	57.8289	63.5263	89.2857	97.0000	83.5132	89.1697	82.6974	-	-	-	-	-	-	-
SST-2	64.4889	60.1651	69.4823	55.7368	57.3421	64.2895	89.2857	98.0000	83.7500	87.0036	82.1579	95.4128	-	-	-	-	-	-
dbpedia	66.0656	59.3051	74.5656	56.1974	57.1053	64.7895	85.7143	95.0000	84.5395	83.7545	82.5526	95.8716	99.0789	-	-	-	-	-
agnews	66.9819	59.1675	77.1746	57.8816	57.6053	64.8421	87.5000	95.0000	84.8421	81.5884	82.5658	95.7569	97.0132	92.8684	-	-	-	-
yahoo	62.8714	55.2116	72.9990	55.1316	51.6842	64.4474	89.2857	94.0000	76.2632	81.2274	81.1447	95.0688	90.2895	89.2105	72.5921	-	-	-
MultiRC	66.9502	61.1627	73.9474	56.6974	56.6053	65.0263	89.2857	96.0000	84.8947	81.5884	81.3947	95.2982	98.3158	79.8289	64.2895	76.1345	-	-
BoolQA	67.4003	61.3691	74.7462	58.5395	58.3553	64.9605	87.5000	97.0000	81.3158	79.4224	82.2105	95.7569	98.5789	82.7105	63.8947	74.8762	84.8624	-
WiC	63.2068	57.2411	70.5607	57.1053	55.1053	57.0395	78.5714	96.0000	68.6316	62.0939	79.2368	87.8440	98.5921	83.8158	64.2368	67.5330	71.1621	74.4514

TABLE 29: The task-wise performance (%) of Baseline^R+TM (w=dy) with BI under full-parameter setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	63.5458	63.3299	63.7632	63.7632	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	62.1264	63.5019	60.8092	60.6316	60.9868	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	61.0796	60.4403	61.7325	52.3158	50.3026	82.5789	-	-	-	-	-	-	-	-	-	-	-	-
CB	61.3035	59.9587	62.7100	54.4474	52.3289	81.2105	82.1429	-	-	-	-	-	-	-	-	-	-	-
COPA	62.8421	60.4403	65.4426	59.0263	58.8816	77.9342	82.1429	93.0000	-	-	-	-	-	-	-	-	-	-
QQP	63.3991	59.8899	67.3452	53.7500	52.4605	78.7368	75.0000	93.0000	84.0395	-	-	-	-	-	-	-	-	-
RTE	62.4633	60.5435	64.5088	57.2500	52.6053	68.0658	83.9286	91.0000	78.7763	87.7256	-	-	-	-	-	-	-	-
IMDB	65.8217	60.2339	72.5522	59.2763	58.8684	79.0921	91.0714	93.0000	82.0526	87.3646	82.5263	-	-	-	-	-	-	-
SST-2	64.0636	60.0963	68.5918	51.9211	52.4605	72.3684	80.3571	94.0000	80.2763	81.9495	82.0395	94.6101	-	-	-	-	-	-
dbpedia	65.4618	59.6491	72.5296	53.1974	53.6579	69.8816	66.0714	92.0000	74.4868	80.1444	81.9079	94.8394	99.0000	-	-	-	-	-
agnews	65.0111	58.0323	73.8978	44.7500	48.8026	72.9737	66.0714	90.0000	77.0526	79.0614	81.8026	93.8073	98.6053	90.6711	-	-	-	-
yahoo	62.6828	55.3492	72.2567	49.1974	49.0132	64.7237	33.9286	84.0000	77.9737	71.1191	80.4342	93.0046	96.4474	87.1842	70.8684	-	-	-
MultiRC	63.8695	56.3811	73.6517	51.2895	49.1053	73.0921	73.2143	88.0000	79.7632	80.5054	79.4605	92.2018	97.0526	87.1974	68.9605	74.7937	-	-
BoolQA	65.6708	58.4795	74.8786	55.5132	53.0000	70.7105	73.2143	90.0000	78.6974	77.2563	80.6974	92.0872	97.0658	88.0526	70.3289	73.7005	82.9358	-
WiC	63.3746	55.7620	73.3944	55.0921	54.5263	69.2632	60.7143	91.0000	72.6316	72.9242	81.4211	93.1193	97.2895	86.9868	70.0395	69.1419	74.2508	71.7868

TABLE 30: The task-wise performance (%) of Baseline under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.8306	65.0843	64.5789	64.5789	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.1428	64.4995	61.8421	62.4605	61.2237	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	62.9071	63.2611	62.5570	55.5000	54.8684	77.3026	-	-	-	-	-	-	-	-	-	-	-	-
CB	62.9278	61.1283	64.8364	56.3026	55.7763	82.2500	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	63.3243	61.6443	65.0984	56.8947	55.5526	82.3026	87.5000	94.0000	-	-	-	-	-	-	-	-	-	-
QQP	64.3577	61.1283	67.9474	57.6711	55.4737	75.2368	75.0000	93.0000	83.0263	-	-	-	-	-	-	-	-	-
RTE	62.9565	59.8555	66.3964	55.5132	53.9605	72.8026	76.7857	90.0000	82.1184	88.4477	-	-	-	-	-	-	-	-
IMDB	64.9734	59.8555	71.0483	56.6447	55.9079	77.3553	76.7857	92.0000	82.8289	88.0866	81.5658	-	-	-	-	-	-	-
SST-2	65.9899	60.2683	72.9118	58.5263	58.2632	80.1184	78.5714	92.0000	82.4079	82.6715	82.0526	95.0688	-	-	-	-	-	-
dbpedia	67.4252	60.1307	76.7338	58.1974	57.7105	79.4474	80.3571	91.0000	81.8684	84.8375	81.9605	94.4954	98.6711	-	-	-	-	-
agnews	67.7679	59.9587	77.9158	59.8289	57.8684	74.7500	67.8571	88.0000	80.7368	83.3935	81.9079	93.6927	97.6842	90.5658	-	-	-	-
yahoo	64.5686	57.3787	73.8185	54.6842	51.0263	65.7105	58.9286	85.0000	80.1184	80.5054	81.6316	92.5459	96.0263	88.4079	70.5132	-	-	-
MultiRC	66.1018	58.6515	75.7203	56.4211	53.9079	76.0000	73.2143	85.0000	81.3421	84.1155	81.8289	93.3486	96.8684	88.4079	69.5132	74.2162	-	-
BoolQA	66.6477	58.8923	76.7555	59.8289	58.0789	79.2763	80.3571	85.0000	77.6447	79.4224	81.7237	93.4633	97.9342	88.8158	70.4474	70.9365	81.0703	-
WiC	66.0296	58.6515	75.5310	60.2105	56.5263	77.1842	71.4286	85.0000	74.0789	79.7834	81.4342	93.5780	97.5789	88.6579	69.2237	71.0809	76.1774	70.6897

TABLE 31: The task-wise performance (%) of Baseline^R under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.0840	64.7747	63.4079	63.4079	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	62.9931	65.2563	60.8816	60.6974	61.0658	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	64.7046	63.9491	65.4781	56.3684	56.9868	83.0789	-	-	-	-	-	-	-	-	-	-	-	-
CB	64.5700	63.9147	65.2389	55.7895	57.3421	82.4079	89.2857	-	-	-	-	-	-	-	-	-	-	-
COPA	63.7832	64.3963	63.1817	57.6974	59.6447	71.6184	83.9286	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	67.3307	65.1875	69.6197	59.8553	56.6842	77.6184	82.1429	97.0000	83.8684	-	-	-	-	-	-	-	-	-
RTE	62.2957	63.7083	60.9444	54.6974	47.0658	60.6579	83.9286	98.0000	79.6711	89.1697	-	-	-	-	-	-	-	-
IMDB	64.6159	64.0179	65.2252	44.5395	48.6184	65.9737	85.7143	97.0000	83.2763	85.5596	82.4079	-	-	-	-	-	-	-
SST-2	66.5814	64.5683	68.7241	55.3816	53.3421	68.1579	85.7143	95.0000	80.7632	80.8664	82.0132	95.2982	-	-	-	-	-	-
dbpedia	67.6828	61.1971	75.7062	60.5395	59.0789	72.0658	80.3571	92.0000	78.6184	81.2274	82.2368	94.9541	99.0395	-	-	-	-	-
agnews	67.5281	60.0275	77.1709	58.6053	56.2500	71.9342	76.7857	92.0000	78.4868	83.0325	82.2763	94.9541	98.6711	91.5263	-	-	-	-
yahoo	63.4914	57.0347	71.5965	52.7105	46.5000	57.9342	37.5000	68.0000	74.7763	76.5343	81.7368	92.7752	96.6316	89.8947	70.2763	-	-	-
MultiRC	64.8842	57.7227	74.0744	54.5132	48.5000	70.5000	82.1429	85.0000	78.7105	81.5884	81.5395	92.7752	97.5526	89.2895	68.7237	75.0825	-	-
BoolQA	65.9588	58.8579	75.0082	58.9079	55.3816	69.1974	80.3571	84.0000	75.3158	77.2563	82.0132	94.9541	98.1711	87.7895	68.6842	73.2261	82.4771	-
WiC	63.8621	56.5187	73.3986	58.3026	56.3816	69.5526	62.5000	84.0000	63.4605	71.8412	81.9737	94.1514	98.1842	88.3026	68.6316	70.4002	78.2263	71.6301

TABLE 32: The task-wise performance (%) of Baseline^R with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.6367	66.0819	65.1974	65.1974	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.7459	65.6003	61.9934	61.8289	62.1579	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.2054	64.3963	66.0351	58.1316	56.4605	83.5132	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.5178	63.9147	67.2034	59.8421	58.2763	83.3553	85.7143	-	-	-	-	-	-	-	-	-	-	-
COPA	66.0331	64.9811	67.1197	61.0658	58.5132	81.2368	89.2857	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.7616	64.6371	69.0306	59.5000	55.6579	76.4737	71.4286	96.0000	84.1184	-	-	-	-	-	-	-	-	-
RTE	65.6742	64.1211	67.3045	58.1974	53.9342	72.6842	82.1429	95.0000	83.1974	87.3646	-	-	-	-	-	-	-	-
IMDB	68.0322	63.6739	73.0310	61.4474	58.2763	78.6842	85.7143	96.0000	83.7237	85.5596	82.1711	-	-	-	-	-	-	-
SST-2	68.0461	63.5707	73.1993	62.2237	57.7237	77.5000	78.5714	96.0000	82.8289	83.7545	82.3553	96.2156	-	-	-	-	-	-
dbpedia	69.5051	63.1235	77.3222	61.6711	58.7763	78.3026	80.3571	93.0000	81.3684	84.4765	82.2895	94.9541	99.0132	-	-	-	-	-
agnews	68.8300	62.0227	77.3158	58.4605	53.8816	74.3947	75.0000	91.0000	80.8553	84.1155	81.9737	93.6927	98.0526	91.3026	-	-	-	-
yahoo	65.4846	59.0987	73.4176	55.3158	49.8026	66.5263	62.5000	86.0000	78.2895	80.5054	81.2632	91.7431	95.4868	88.2895	69.9211	-	-	-
MultiRC	66.7758	59.9243	75.3962	56.0000	51.9342	75.0263	71.4286	86.0000	81.4474	85.9206	81.7632	92.6606	97.1053	87.6579	69.7500	75.4125	-	-
BoolQA	66.9228	59.8555	75.8825	59.5789	56.7368	76.4868	89.2857	82.0000	75.7632	81.9495	81.8684	93.2339	97.6053	85.9211	69.8158	72.3391	83.2110	-
WiC	67.2289	61.0251	74.8367	59.2632	55.7895	74.1579	80.3571	85.0000	72.2895	76.1733	81.9474	92.6606	97.6711	86.6447	69.6184	70.7715	79.2966	71.4734

TABLE 33: The task-wise performance (%) of Baseline^R+KL under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.5996	65.7379	63.5000	63.5000	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.6099	65.1531	62.1382	62.5263	61.7500	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	64.9267	64.7059	65.1491	56.4737	56.0526	82.9211	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.0925	64.9467	65.2389	56.9342	56.8816	81.6974	92.8571	-	-	-	-	-	-	-	-	-	-	-
COPA	65.4596	66.4603	64.4886	59.4474	60.4474	72.9605	87.5000	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.9799	65.5315	68.4939	54.2500	55.7895	79.6579	83.9286	99.0000	83.7632	-	-	-	-	-	-	-	-	-
RTE	65.7386	65.1875	66.2991	56.2237	54.4474	71.5526	85.7143	97.0000	81.6842	86.6426	-	-	-	-	-	-	-	-
IMDB	64.6539	64.8435	64.4654	42.0526	40.4737	73.4474	85.7143	94.0000	82.4474	84.8375	82.6184	-	-	-	-	-	-	-
SST-2	68.2430	65.6347	71.0673	59.6447	54.6579	72.8947	85.7143	97.0000	81.9868	80.8664	82.5789	95.1835	-	-	-	-	-	-
dbpedia	71.2101	65.9099	77.4374	63.0658	58.4737	77.1842	82.1429	96.0000	81.6974	81.5884	82.8026	95.1835	98.9342	-	-	-	-	-
agnews	70.7100	65.2907	77.1104	58.8026	48.8158	76.5921	85.7143	95.0000	81.8553	80.1444	82.5526	94.9541	97.9868	90.7105	-	-	-	-
yahoo	66.9993	61.2315	73.9667	56.6974	51.2105	70.1184	80.3571	91.0000	76.6842	76.1733	80.9474	93.8073	95.8816	86.6316	70.9342	-	-	-
MultiRC	68.4132	62.5043	75.5560	59.0395	52.6053	75.9474	82.1429	89.0000	79.9737	77.9783	81.3289	92.5459	97.2237	85.8158	70.2368	75.5776	-	-
BoolQA	68.7035	63.0547	75.4639	58.8158	53.3421	72.0658	87.5000	91.0000	77.5395	74.0072	82.0526	94.1514	97.0263	86.4474	71.3553	74.1130	83.7003	-
WiC	66.7872	61.5755	72.9626	57.1447	50.4079	67.3158	75.0000	92.0000	68.8289	68.5921	82.1711	93.5780	96.3684	86.2632	70.2632	72.3391	80.0000	71.0031

TABLE 34: The task-wise performance (%) of Baseline^R+KL with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.4596	66.2539	64.6842	64.6842	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.1957	66.1851	62.3224	62.8684	61.7763	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.3919	65.6347	67.1667	59.5789	58.0132	83.9079	-	-	-	-	-	-	-	-	-	-	-	-
CB	67.2246	66.2883	68.1878	63.7237	60.9211	79.7368	92.8571	-	-	-	-	-	-	-	-	-	-	-
COPA	66.6241	66.4259	66.8235	64.6842	60.9737	74.2763	89.2857	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	68.4241	66.3915	70.5852	62.7632	58.2632	77.2500	83.9286	97.0000	83.6184	-	-	-	-	-	-	-	-	-
RTE	67.2146	65.4283	69.1013	60.9474	55.7500	75.7895	85.7143	92.0000	82.7895	88.4477	-	-	-	-	-	-	-	-
IMDB	68.7426	65.2563	72.6225	61.3816	59.2368	77.1053	83.9286	95.0000	82.3421	86.2816	82.1711	-	-	-	-	-	-	-
SST-2	68.7234	65.6003	72.1588	61.7895	57.4211	74.2500	82.1429	95.0000	81.7105	85.9206	82.0132	95.9862	-	-	-	-	-	-
dbpedia	69.5023	64.2243	75.7254	62.3421	60.6316	71.1974	58.9286	94.0000	76.3947	83.7545	82.1053	95.2982	99.0263	-	-	-	-	-
agnews	68.7413	62.3667	76.5673	59.8947	55.8289	67.9605	62.5000	92.0000	77.8684	85.5596	81.9737	93.1193	98.6316	91.4868	-	-	-	-
yahoo	64.9156	59.2363	71.7994	53.9868	50.5395	55.8553	39.2857	88.0000	74.1579	84.8375	80.5921	91.7431	97.7500	88.7895	69.9868	-	-	-
MultiRC	67.4783	61.1627	75.2483	54.7895	52.3947	74.0789	75.0000	81.0000	79.7105	85.5596	81.0658	91.3991	98.3947	88.7632	70.6842	74.9381	-	-
BoolQA	68.3416	61.9195	76.2499	60.0000	58.1579	73.6316	87.5000	87.0000	75.8026	80.8664	81.8947	94.4954	98.5789	87.6842	70.6316	73.2880	83.2722	-
WiC	66.9565	61.5411	73.4170	57.6184	54.9737	70.9211	83.9286	85.0000	65.8684	76.1733	80.9605	93.6927	98.6053	87.7500	69.5658	68.3375	77.6453	71.4734

TABLE 35: The task-wise performance (%) of Baseline^R+L1 under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.7243	65.8411	63.6447	63.6447	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.0712	66.1507	62.1184	62.5000	61.7368	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.7418	66.0131	67.4868	58.8158	60.5789	83.0658	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.1626	65.5659	66.7702	59.2763	60.6579	80.1974	91.0714	-	-	-	-	-	-	-	-	-	-	-
COPA	65.0898	66.3571	63.8700	60.0395	59.8816	71.0395	91.0714	98.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.9136	65.9787	67.8754	56.5000	55.9474	76.0658	71.4286	94.0000	82.6184	-	-	-	-	-	-	-	-	-
RTE	64.9409	65.6003	64.2947	59.5921	57.1974	61.9342	80.3571	95.0000	77.0526	88.4477	-	-	-	-	-	-	-	-
IMDB	67.4378	65.8755	69.0761	58.8289	56.0526	68.0395	91.0714	92.0000	78.6711	87.3646	82.6579	-	-	-	-	-	-	-
SST-2	68.4099	66.0475	70.9477	60.5132	56.5132	71.0658	89.2857	93.0000	80.6184	85.1986	82.3289	94.9541	-	-	-	-	-	-
dbpedia	70.7552	66.1163	76.0942	62.1842	58.7237	70.9079	69.6429	97.0000	80.4868	83.3935	82.5000	95.2982	99.0658	-	-	-	-	-
agnews	71.2588	66.2883	77.0351	59.8684	52.7105	72.6053	83.9286	94.0000	80.1974	80.8664	82.1184	93.3486	98.0132	91.4474	-	-	-	-
yahoo	67.2247	63.4675	71.4548	55.3553	48.9605	56.1447	50.0000	90.0000	75.2763	78.3394	79.2368	80.6193	95.8026	88.3421	71.1316	-	-	-
MultiRC	68.8897	64.2587	74.2401	56.8158	51.0000	70.5921	82.1429	93.0000	80.4079	76.1733	79.4342	84.8624	95.3684	86.7763	70.5789	76.3614	-	-
BoolQA	69.9449	65.0155	75.6832	60.0658	56.2368	70.5263	73.2143	92.0000	77.5132	74.3682	80.9211	92.6606	95.8289	88.8684	70.5526	74.8556	83.5474	-
WiC	68.9253	65.1875	73.1178	57.2632	52.7632	65.3553	69.6429	86.0000	69.4211	69.6751	81.7105	93.0046	97.4079	87.8553	68.9737	72.7517	78.3486	71.6301

TABLE 36: The task-wise performance (%) of Baseline^R+L1 with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.7792	66.4947	65.0789	65.0789	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.9428	65.9443	62.0592	61.5658	62.5526	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	65.7765	64.5683	67.0307	58.3553	59.2632	83.4737	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.2660	64.1899	68.4809	61.0789	61.1316	83.0921	87.5000	-	-	-	-	-	-	-	-	-	-	-
COPA	66.9421	65.8067	68.1173	62.6184	61.4868	79.7368	87.5000	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	67.6247	65.7379	69.6230	62.9605	59.1447	73.0263	78.5714	92.0000	83.0000	-	-	-	-	-	-	-	-	-
RTE	66.5499	64.5339	68.6959	60.9868	57.4079	72.4211	85.7143	92.0000	82.8947	86.2816	-	-	-	-	-	-	-	-
IMDB	67.4151	63.9835	71.2357	60.0263	57.5526	73.3289	85.7143	94.0000	82.5789	85.1986	81.7763	-	-	-	-	-	-	-
SST-2	67.4642	63.9147	71.4311	59.9211	55.5132	74.6974	82.1429	93.0000	81.9868	84.8375	81.4211	95.5275	-	-	-	-	-	-
dbpedia	70.4538	64.5339	77.5696	63.4868	60.5921	77.1316	80.3571	94.0000	80.7368	82.3105	82.1711	95.0688	98.8816	-	-	-	-	-
agnews	69.4072	62.9859	77.2865	58.9868	53.8553	73.8553	83.9286	95.0000	80.6447	83.3935	82.1184	93.9220	97.8947	91.2368	-	-	-	-
yahoo	65.5182	58.2043	74.9344	56.9737	53.3158	69.7237	69.6429	83.0000	80.0526	83.0325	81.5263	93.4633	96.8026	88.3816	70.2105	-	-	-
MultiRC	67.6884	60.8875	76.1997	59.1711	55.5921	73.3947	83.9286	87.0000	80.5789	84.4765	81.7895	93.6927	97.5000	88.6711	70.8684	75.4538	-	-
BoolQA	68.5123	62.0571	76.4664	60.3947	56.9211	74.1842	89.2857	90.0000	79.3553	82.3105	81.8553	93.6927	97.7368	87.6053	70.0395	73.9068	82.9969	-
WiC	67.6656	61.6787	74.9397	57.3289	54.4605	74.2500	83.9286	87.0000	76.5263	77.2563	81.6579	93.3486	98.3158	86.0526	69.7368	71.6378	77.8287	70.5329

TABLE 37: The task-wise performance (%) of Baseline^R+L2 under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.5011	65.2563	63.7632	63.7632	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.3512	65.8067	61.0724	60.2237	61.9211	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	64.9711	65.4627	64.4868	54.7895	56.6711	82.0000	-	-	-	-	-	-	-	-	-	-	-	-
CB	65.8420	65.2907	66.4027	57.3816	59.6447	81.9868	92.8571	-	-	-	-	-	-	-	-	-	-	-
COPA	65.4726	65.3939	65.5515	58.7237	60.4605	76.8816	91.0714	96.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.8773	66.0819	67.6921	53.9474	55.3816	77.3684	78.5714	93.0000	83.6579	-	-	-	-	-	-	-	-	-
RTE	66.1048	65.5659	66.6526	58.2368	55.0395	70.1711	89.2857	98.0000	81.8816	85.9206	-	-	-	-	-	-	-	-
IMDB	66.9156	65.4971	68.3970	53.5658	47.6447	73.4079	91.0714	94.0000	83.7368	85.1986	82.5132	-	-	-	-	-	-	-
SST-2	67.7855	65.7035	70.0038	56.9868	53.4211	71.1447	87.5000	95.0000	82.4474	80.1444	82.2632	95.5275	-	-	-	-	-	-
dbpedia	70.9434	65.6691	77.1389	61.4605	58.1184	77.0263	80.3571	97.0000	82.1053	83.0325	82.6447	95.0688	98.9211	-	-	-	-	-
agnews	70.1363	64.4307	76.9507	57.8158	52.6447	73.0132	80.3571	91.0000	81.6579	84.8375	82.3684	93.4633	97.1711	91.5921	-	-	-	-
yahoo	68.1250	63.3643	73.6591	54.1184	52.6974	63.7763	71.4286	90.0000	78.4605	77.2563	81.6974	91.7431	97.3816	89.5000	69.2368	-	-	-
MultiRC	67.9842	63.4675	73.1931	50.9868	48.1184	72.8158	75.0000	93.0000	79.9211	75.4513	80.8289	91.2844	91.7500	88.5658	68.9079	75.1031	-	-
BoolQA	69.8028	64.2587	76.3938	57.9868	70.4474	75.0000	92.0000	78.3026	76.5343	82.2237	94.1514	97.8553	89.9605	71.2763	73.7417	82.8746	-	-
WiC	68.9476	64.4651	74.1000	57.0921	55.8553	66.1974	64.2857	90.0000	69.8816	67.5090	82.0263	93.0046	97.3026	90.0789	70.3553	72.9167	79.9694	76.1755

TABLE 38: The task-wise performance (%) of Baseline^R+L2 with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.9802	66.0131	65.9474	65.9474	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	64.2013	66.2195	62.3026	62.9868	61.6184	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.7877	66.3915	67.1886	59.8553	58.2105	83.5000	-	-	-	-	-	-	-	-	-	-	-	-
CB	67.1974	66.6323	67.7721	61.9079	60.1447	81.1184	87.5000	-	-	-	-	-	-	-	-	-	-	-
COPA	66.5393	66.4947	66.5839	63.3553	59.4079	76.4605	89.2857	94.0000	-	-	-	-	-	-	-	-	-	-
QQP	68.0480	65.9787	70.2513	61.8684	57.3158	77.2237	80.3571	93.0000	84.2237	-	-	-	-	-	-	-	-	-
RTE	66.7942	65.7379	67.8851	59.5921	54.1447	73.0789	89.2857	92.0000	83.4737	89.1697	-	-	-	-	-	-	-	-
IMDB	68.4342	65.9787	71.0795	59.2763	56.1711	74.0921	83.9286	95.0000	82.5526	87.3646	82.3026	-	-	-	-	-	-	-
SST-2	67.9225	66.0475	69.9071	59.2632	53.1711	70.8026	82.1429	96.0000	80.2500	81.9495	82.1711	96.1009	-	-	-	-	-	-
dbpedia	69.6229	65.1531	74.7511	61.8421	57.1184	70.9342	82.1429	96.0000	74.2368	82.6715	82.5789	95.5275	98.7895	-	-	-	-	-
agnews	70.5449	64.9811	77.1507	60.7500	54.9737	70.6316	73.2143	97.0000	79.6711	83.3935	82.6184	94.9541	98.2500	90.6579	-	-	-	-
yahoo	68.2991	64.3619	72.7494	57.3684	49.8026	60.7632	58.9286	86.0000	72.1711	78.7004	81.8553	91.6284	97.1842	89.3026	71.0921	-	-	-
MultiRC	70.4566	65.1531	76.7001	61.8553	56.3816	75.4342	80.3571	88.0000	79.4474	78.3394	81.7895	91.9725	97.8816	87.8816	71.1447	76.3820	-	-
BoolQA	69.9710	65.4627	75.1463	61.4605	57.3289	70.0526	78.5714	89.0000	70.1447	80.1444	81.8421	94.7248	97.4211	88.4737	68.6974	74.3193	83.6086	-
WiC	69.8567	65.3251	75.0639	59.3026	54.5526	71.9737	83.9286	89.0000	74.2368	75.8123	82.0921	93.4633	97.6053	88.3026	67.8421	74.2987	81.9572	71.9436

TABLE 39: The task-wise performance (%) of Baseline^R+TM under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.7595	66.0131	63.5526	63.5526	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.1847	65.7379	60.8224	60.8947	60.7500	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	66.0735	66.1163	66.0307	57.2237	57.9868	82.8816	-	-	-	-	-	-	-	-	-	-	-	-
CB	66.4132	66.2539	66.5733	59.7105	59.9474	79.8421	96.4286	-	-	-	-	-	-	-	-	-	-	-
COPA	65.8838	66.2195	65.5515	61.7368	61.2763	73.0526	92.8571	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	67.7531	66.4947	69.0601	61.8421	58.5789	71.6974	87.5000	97.0000	83.6184	-	-	-	-	-	-	-	-	-
RTE	65.7343	66.4259	65.0569	59.5132	56.1711	61.5526	80.3571	95.0000	81.6053	89.1697	-	-	-	-	-	-	-	-
IMDB	62.0098	66.1851	58.3301	31.9079	28.3158	65.1711	85.7143	93.0000	82.5921	87.3646	81.9474	-	-	-	-	-	-	-
SST-2	68.8088	66.5979	71.1716	61.4211	56.4342	69.6842	85.7143	93.0000	82.1711	83.7545	82.5132	95.4128	-	-	-	-	-	-
dbpedia	71.3194	66.3227	77.1304	61.8026	59.6842	76.1184	85.7143	93.0000	80.5054	82.6711	95.7569	98.9737	-	-	-	-	-	-
agnews	71.7573	66.5291	77.8773	59.7763	57.2895	73.9737	87.5000	96.0000	80.5921	78.3394	82.3684	95.1835	98.4342	90.3947	-	-	-	-
yahoo	68.5125	64.5683	72.9700	56.5789	53.5658	55.8816	41.0714	92.0000	78.0395	71.1191	81.8553	93.4633	96.8421	88.5263	70.1711	-	-	-
MultiRC	69.0975	64.8779	73.9041	54.7763	52.9868	67.8158	71.4286	92.0000	77.5789	74.3682	81.4474	90.4817	96.3026	87.9211	69.0526	75.8045	-	-
BoolQA	69.8066	65.1187	75.2218	58.9211	56.2500	68.4342	73.2143	91.0000	75.5921	69.3141	81.6447	93.9220	97.6447	88.5789	69.4605	74.2781	83.8838	-
WiC	68.4606	64.6371	72.7650	56.0263	51.4868	64.6974	73.2143	90.0000	70.3158	66.0650	80.7895	91.3991	97.5263	87.7632	68.8158	72.8754	78.8991	71.1599

TABLE 40: The task-wise performance (%) of Baseline^R+TM with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.6995	66.1163	63.3421	63.3421	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	62.9135	66.7011	59.5329	59.7895	59.2763	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	61.3383	66.3915	57.0000	49.3947	49.5000	72.1053	-	-	-	-	-	-	-	-	-	-	-	-
CB	60.6811	66.3227	55.9240	50.6184	49.8289	67.2237	69.6429	-	-	-	-	-	-	-	-	-	-	-
COPA	59.1370	66.4947	53.2453	51.1842	48.9868	59.1184	57.1429	85.0000	-	-	-	-	-	-	-	-	-	-
QQP	61.1059	66.0819	56.8268	49.3816	48.7763	50.5263	35.7143	75.0000	78.5395	-	-	-	-	-	-	-	-	-
RTE	62.2074	66.6323	58.3336	49.3289	47.7237	58.0921	48.2143	80.0000	77.2895	77.2563	-	-	-	-	-	-	-	-
IMDB	64.8620	66.4603	63.3388	49.6184	50.8947	57.5132	48.2143	57.0000	76.3421	73.2852	82.1579	-	-	-	-	-	-	-
SST-2	65.5385	66.7355	64.3837	54.5263	51.6842	54.9868	58.9286	86.0000	75.1316	65.7040	81.8553	94.3807	-	-	-	-	-	-
dbpedia	67.1732	66.3915	67.9736	59.6842	52.1579	44.7895	39.2857	84.0000	68.1579	61.0108	81.7237	93.4633	98.6579	-	-	-	-	-
agnews	67.6305	66.3915	68.9166	51.5526	47.2500	42.0789	28.5714	78.0000	69.5789	63.8989	81.6053	93.0046	97.7500	90.1974	-	-	-	-
yahoo	64.8817	65.8411	63.9498	47.6711	43.5789	43.1316	46.4286	33.0000	64.2632	68.5921	79.7763	85.5505	89.0395	70.9605	71.0658	-	-	-
MultiRC	63.2124	66.1163	60.5529	29.1579	26.2632	58.1711	50.0000	1.0000	70.6579	64.9819	79.9737	87.7294	85.1974	59.5263	66.1711	71.3490	-	-
BoolQA	65.5736	66.2195	64.9403	54.2763	49.8158	49.2105	37.5000	82.0000	65.4211	63.8989	80.1316	92.2018	91.6579	66.7368	59.7632	60.1279	70.6728	-
WiC	64.9147	65.9443	63.9167	53.1711	50.0658	41.4342	37.5000	84.0000	73.8553	67.8700	79.8553	91.3991	91.2237	69.0921	49.6579	60.6848	68.2263	61.7555

TABLE 41: The task-wise performance (%) of Baseline^R+TM (w=100) under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.8046	66.5979	63.1053	63.1053	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.3197	66.4947	60.4342	61.8553	59.0132	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	63.4808	66.6323	60.6140	51.7632	49.7105	80.3684	-	-	-	-	-	-	-	-	-	-	-	-
CB	60.6909	66.4947	55.8190	49.1447	46.2895	71.8421	80.3571	-	-	-	-	-	-	-	-	-	-	-
COPA	54.3829	66.2195	46.1361	47.3553	42.0132	48.2632	64.2857	95.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.0703	66.8387	65.3194	56.5658	51.9079	69.6184	76.7857	91.0000	82.7632	-	-	-	-	-	-	-	-	-
RTE	63.2571	66.9763	59.9293	50.2763	47.3026	60.3026	80.3571	89.0000	80.4474	83.3935	-	-	-	-	-	-	-	-
IMDB	66.3707	66.2883	66.4533	54.1053	52.0395	65.3026	76.7857	86.0000	78.1579	79.0614	81.8684	-	-	-	-	-	-	-
SST-2	67.3507	66.4259	68.3017	58.3026	55.3947	65.7895	71.4286	91.0000	76.5263	72.9242	81.9079	95.2982	-	-	-	-	-	-
dbpedia	68.0776	66.1163	70.1588	57.4737	51.6711	57.6447	50.0000	91.0000	70.7368	63.8989	82.0263	93.5780	98.8158	-	-	-	-	-
agnews	67.9356	66.3915	69.5533	47.7895	42.9605	49.9474	42.8571	86.0000	74.0000	68.2310	81.2237	93.3486	97.7763	90.4737	-	-	-	-
yahoo	66.1892	65.9443	66.4359	47.7237	43.6184	46.4079	46.4286	61.0000	66.6053	68.5921	80.3026	90.5963	89.8158	83.8289	70.5526	-	-	-
MultiRC	66.9679	66.0131	67.9507	47.4474	43.1316	64.1974	69.6429	59.0000	73.7632	72.2022	79.8684	87.0413	82.0789	77.5000	69.1974	74.5050	-	-
BoolQA	67.4027	66.0131	68.8521	53.1316	49.2105	61.8421	60.7143	38.0000	72.0132	69.6751	81.3684	93.1193	94.1579	67.7105	64.2763	68.3375	80.6728	-
WiC	67.1323	65.5659	68.7755	53.7500	49.8947	55.1711	58.9286	59.0000	70.7632	66.7870	81.2895	91.9725	95.0921	80.7237	60.5263	66.6048	73.2110	69.7492

TABLE 42: The task-wise performance (%) of Baseline^R+TM (w=100) with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.8175	66.2195	63.4737	63.4737	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.4872	66.8731	60.4276	60.8158	60.0395	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	63.1611	66.3571	60.2588	52.3158	51.3947	77.0658	-	-	-	-	-	-	-	-	-	-	-	-
CB	61.1721	66.4603	56.6635	50.3553	49.4605	70.0526	73.2143	-	-	-	-	-	-	-	-	-	-	-
COPA	60.4614	66.5635	55.3842	52.5132	49.6053	63.6711	57.1429	82.0000	-	-	-	-	-	-	-	-	-	-
QQP	61.7453	66.4259	57.6810	49.7105	49.2368	51.8421	37.5000	78.0000	79.8158	-	-	-	-	-	-	-	-	-
RTE	61.6476	66.6667	57.3314	48.0789	47.2368	56.0395	39.2857	83.0000	76.9737	79.0614	-	-	-	-	-	-	-	-
IMDB	64.9994	66.2539	63.7915	45.4211	49.7763	65.3026	46.4286	30.0000	76.5132	71.8412	82.2237	-	-	-	-	-	-	-
SST-2	66.2760	66.6667	65.8898	54.5789	53.0921	60.8158	51.7857	88.0000	75.5789	70.7581	81.7763	94.1514	-	-	-	-	-	-
dbpedia	67.3716	66.4603	68.3083	59.1974	51.4342	45.0789	33.9286	88.0000	71.2895	64.6209	81.2632	93.5780	98.8158	-	-	-	-	-
agnews	68.2192	66.6323	69.8835	52.4605	49.1316	44.1053	37.5000	89.0000	70.9737	66.0650	81.6316	92.4312	98.0921	90.3289	-	-	-	-
yahoo	65.5524	66.1163	64.9980	49.4079	43.6974	46.9737	57.1429	79.0000	63.9474	64.6209	79.5789	89.3349	88.0132	74.3816	71.0789	-	-	-
MultiRC	65.5709	66.1851	64.9680	34.4474	43.2368	60.0526	53.5714	70.0000	72.4737	67.8700	80.5658	89.6789	90.8816	64.9605	66.5395	70.7096	-	-
BoolQA	65.8838	66.3571	65.4173	53.1316	49.0526	49.8553	48.2143	85.0000	66.0526	66.0650	80.8947	92.6606	92.4079	68.4211	59.2763	62.4587	72.0489	-
WiC	65.3352	66.2539	64.4417	51.6711	49.8158	43.2895	35.7143	83.0000	74.0132	68.5921	80.5000	91.8578	91.3816	70.9605	49.1974	62.6238	69.6942	67.7116

TABLE 43: The task-wise performance (%) of Baseline^R+TM (w=dynamic) under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	65.2051	66.8731	63.6184	63.6184	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	63.6777	66.9419	60.7171	61.3026	60.1316	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	63.6676	66.5635	61.0132	52.0395	50.0526	80.9474	-	-	-	-	-	-	-	-	-	-	-	-
CB	60.9491	66.1507	56.5060	50.0395	47.2237	72.0526	83.9286	-	-	-	-	-	-	-	-	-	-	-
COPA	59.1730	66.0819	53.5721	52.8421	49.6316	57.6447	67.8571	91.0000	-	-	-	-	-	-	-	-	-	-
QQP	66.5630	66.7011	66.4256	58.6711	54.7500	68.6579	82.1429	94.0000	83.1447	-	-	-	-	-	-	-	-	-
RTE	64.3032	66.8387	61.9531	51.6053	50.1053	64.5658	82.1429	94.0000	80.0789	86.2816	-	-	-	-	-	-	-	-
IMDB	65.4650	66.0131	64.9260	53.7368	45.4474	64.6184	80.3571	93.0000	77.8026	81.2274	81.9474	-	-	-	-	-	-	-
SST-2	66.7806	65.6691	67.9303	57.4868	54.3026	65.9211	85.7143	95.0000	75.7237	77.2563	82.2500	95.2982	-	-	-	-	-	-
dbpedia	68.2080	66.3571	70.1652	57.0000	51.9474	57.8553	58.9286	83.0000	71.4474	58.4838	81.8158	92.8899	98.6579	-	-	-	-	-
agnews	68.3889	66.0131	70.9421	50.2105	47.2500	56.0263	66.0714	81.0000	71.8026	64.2599	81.0000	90.5963	98.0921	90.1053	-	-	-	-
yahoo	66.3217	65.7723	66.8803	51.7895	45.6974	43.2237	35.7143	84.0000	64.4474	63.1769	79.4079	86.2385	91.4342	85.6579	71.3026	-	-	-
MultiRC	67.7357	65.4971	70.1328	50.1316	45.8816	64.2763	82.1429	86.0000	76.9079	74.7292	78.8816	89.1055	88.3947	81.5658	69.0921	75.2888	-	-
BoolQA	68.0276	65.8411	70.3644	55.1447	52.8553	59.5921	67.8571	58.0000	74.7895	68.9531	81.6316	93.2339	94.1974	71.6579	66.6184	68.6881	82.2324	-
WiC	66.5906	65.4627	67.7580	52.6579	50.3947	54.6184	66.0714	57.0000	65.4605	63.1769	81.2763	91.9725	94.1184	77.0789	61.5526	64.7690	76.9419	72.5705

TABLE 44: The task-wise performance (%) of Baseline^R+TM (w=dynamic) with BI under LoRA setting

Curr.Task	F1 Avg	MMLU	Tasks Avg	yelp	amazon	MNLI	CB	COPA	QQP	RTE	IMDB	SST-2	dbpedia	agnews	yahoo	MultiRC	BoolQA	WiC
yelp	64.3813	64.7403	64.0263	64.0263	-	-	-	-	-	-	-	-	-	-	-	-	-	-
amazon	61.3441	62.4011	60.3224	59.9079	60.7368	-	-	-	-	-	-	-	-	-	-	-	-	-
MNLI	63.4462	61.0939	65.9868	58.4737	58.6579	80.8289	-	-	-	-	-	-	-	-	-	-	-	-
CB	63.4463	61.0939	65.9870	59.3158	59.6842	78.8421	82.1429	-	-	-	-	-	-	-	-	-	-	-
COPA	64.0736	62.2291	66.0307	59.9211	59.9605	77.7500	80.3571	93.0000	-	-	-	-	-	-	-	-	-	-
QQP	65.5306	61.7819	69.7637	59.6316	59.0395	78.2368	73.2143	91.0000	81.8421	-	-	-	-	-	-	-	-	-
RTE	63.8376	61.5411	66.3121	59.3158	59.2368	64.8421	76.7857	91.0000	80.7237	86.2816	-	-	-	-	-	-	-	-
IMDB	65.3259	61.5411	69.6068	60.7368	58.5395	64.4868	76.7857	90.0000	81.4211	87.0036	81.8947	-	-	-	-	-	-	-
SST-2	64.9790	60.2339	70.5356	61.2895	58.1447	65.6711	78.5714	92.0000	82.0263	83.3935	81.9605	94.7248	-	-	-	-	-	-
dbpedia	66.0818	59.1331	74.8811	60.0000	58.2500	66.0921	80.3571	93.0000	80.8684	84.1155	82.2763	94.3807	98.9474	-	-	-	-	-
agnews	65.9922	58.0323	76.4829	60.3421	57.1842	64.7105	80.3571	92.0000	79.6579	84.1155	82.0132	93.9220	98.6316	90.3289	-	-	-	-
yahoo	64.1233	55.9684	75.0600	54.8947	54.4211	69.5395	76.7857	90.0000	79.5000	79.7834	81.9868	92.8899	97.6711	88.7895	71.2500	-	-	-
MultiRC	63.6088	55.3492	74.7659	55.1579	54.9342	68.5658	75.0000	88.0000	80.6711	77.9783	81.5921	92.8899	97.8947	88.0921	70.2500	72.5660	-	-
BoolQA	64.3424	56.0716	75.4753	56.5395	57.3816	72.8947	82.1429	92.0000	77.6053	74.3682	81.9868	93.5780	97.0658	88.5921	70.1711	71.5553	79.5719	-
WiC	63.5707	55.8996	73.6823	56.0921	56.3289	72.2763	66.0714	91.0000	67.2895	71.1191	81.8026	93.0046	97.3289	87.5132	70.5395	69.0388	76.7584	69.2790

TABLE 45: The task-wise performance (%) of O-LoRA